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Search for New Physics in anomalous jets using jet substructures with the ATLAS detector and LHC Run 3 data

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Introduction

It was 4 July 2012 when Joseph Incandela, CMS spokesperson, affirmed: "*They combine to give us a combined significance of five standard deviations*". With these exact words, he announced, confirmed by ATLAS spokesperson Fabiola Gianotti a few minutes later, the first observation of the Higgs boson [1, 2], which completed the Standard Model (SM) framework. Fourteen years later, no comparable discovery has followed within the SM paradigm at collider experiments, strengthening the hypothesis that it is the most complete and well-tested description of nature.
In reality, the SM probably is not as complete as it appears. In fact, there are many phenomena not included in the framework but confirmed by astrophysical and cosmological observations: among these, we can find the neutrino mass measurement, Dark Matter [3] and baryon asymmetry. All these processes fall into the category of New Physics (NP) Beyond the Standard Model (BSM). They have been hunted by ATLAS and CMS experiments using traditional data analysis techniques, targeting specific theoretical models but, unfortunately, nothing has been observed yet and only upper limits on production cross-sections were set. There are many reasons for this unexpected outcome: wrong physical model for experiments, wrong triggering while selecting data, or deviations too small to be observed through normal techniques.
A possible solution to some of these issues could be Anomaly Detection (AD), that is the search for data deviating from what is already known. In our field of interest, Particle Physics, we apply this method to look for anomalies (signal data), deviating from well known reacting and decaying particles (background data). The strength of this method lies in its model agnostic nature, meaning that it does not rely on a specific theoretical model to analyse data. This allows us to explore a more vast range of events, increasing the background rejection efficiency, giving us a first hint of the region where to look more in depth for New Physics with precise data analysis techniques. In this process, unsupervised Machine Learning (ML) helps speeding up everything.
This thesis investigates whether such an unsupervised approach can be applied to a concrete search. Using ATLAS official Monte Carlo (MC) samples simulating the ATLAS detector during Run-3, a search for BSM resonant signals decaying into hadronic final states with large-radius jets (used to probe boosted resonance topologies) has been conducted, preparing the necessary tools to finalise the analysis on real ATLAS data. Leveraging the Anomaly Score (AS) distributions computed by a Graph Neural Network (GNN) for each jet, the analysis strategy was defined, starting with the definition of Working Points (WPs), thresholds on the AS distribution corresponding to a given background rejection fraction. A crucial part of

the strategy was determining the correct modelling function for the background and signals di-jet invariant mass (m_{JJ}) distribution. After tuning and validating a Double-Sided Crystal Ball (DSCB) function to model the signals and a 5-parameter function for the background distribution, fits to the MC samples were executed, deriving the expected upper limits at 95% Confidence Level on m_{JJ} using the CLs method, at different AS WPs. The study on MC samples was concluded and fully validated by performing sensitivity studies based on a binned maximum likelihood fit to the di-jet invariant mass distribution.
The thesis is structured as follows. In Chapter 1 the Standard Model is presented, together with theories of Physics beyond it. The LHC collider and the ATLAS experiment are discussed in Chapter 2. Chapter 4 presents the Anomaly Detection approach. The typical workflow of a Particle Physics analysis is described in Chapter **??**. Chapter 5 presents the analysis conducted during this study and its results. Finally, conclusions are drawn in Chapter 6.

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Chapter 1

(Beyond) The Standard Model

The Standard Model (SM) of Particle Physics is a relativistic Quantum Field Theory (QFT) describing three out of four fundamental interactions, strong, weak and electromagnetic (EM), and the behavior of elementary particles under them. For a long time attempts were made to to merge the fourth fundamental interaction, gravity, with the SM framework, but those have not been validated yet by experimental observations. Nevertheless, for our purposes, gravity does not play a significant role since the energy scales involved in the processes at a collider experiment, as the LHC, are much lower than the Planck scale, where gravitational effects become relevant.
What is remarkable about the SM is that it is a very well tested theory, with predictions confirmed by experimental observations with a very high precision. The discovery of the Higgs boson in 2012 at the LHC, which completed the SM framework, is a clear example of this.
Despite the success of the SM, there are many phenomena that are not explained by it, but are confirmed by astrophysical and cosmological observations, opening the road for Physics Beyond the Standard Model.
This chapter presents a brief overview of the SM, highlighting its major features. In addition, hints for Physics Beyond it are discussed.

1.1 An overview of the Standard Model

The fundamental constituents of the Universe are precisely described by the SM. They are called elementary particles, which are mathematically described as quantum fields $\psi(x)$ and their dynamics are modelled by Lagrangian densities $\mathcal{L}(\psi, \partial_\mu \psi)$ [4]. Such particles can be graphically pictured as presented in Fig. 1.1. They are divided into two fundamental classes based on their spin quantum number. The first class is composed by *fermions*, which cover the first three column of Fig. 1.1 and are organised in three *generations* of increasing proper mass; they have spin 1/2 and constitute *matter*. The second class is that of *bosons*, with integer spin, which *mediate* fundamental interactions and generate mass.
Fermions are further classified into two distinct categories depending on their participation in the strong interaction. The first category is relative to *quarks*. They carry *colour charge* (red, green, blue), making them triplets under $SU(3)$ and therefore interact via all fundamental forces, among which the strong one. Quarks

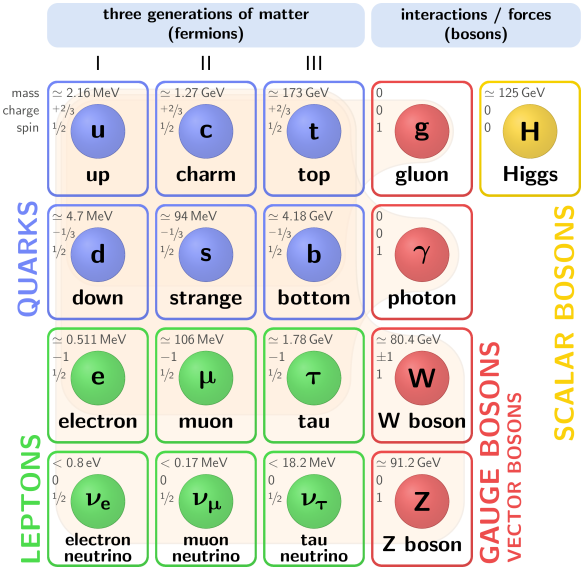


Figure 1.1 The SM framework. Figure from Ref. [5].

exist in six flavors, organized into three generations of weak SU(2) doublets:

$$\begin{pmatrix} u \\ d \end{pmatrix}, \quad \begin{pmatrix} c \\ s \end{pmatrix}, \quad \begin{pmatrix} t \\ b \end{pmatrix} \tag{1.1}$$

Quarks carry a fractional electromagnetic charge: the *up-type* ones (u, c, t) have a charge of $+2/3$, while the *down-type* ones (d, s, b) carry a $-1/3$ charge.
The second category of fermions is associated with *leptons*; they are six of them and similarly grouped into three generations:

$$\begin{pmatrix} e^- \\ \nu_e \end{pmatrix}, \quad \begin{pmatrix} \mu^- \\ \nu_\mu \end{pmatrix}, \quad \begin{pmatrix} \tau^- \\ \nu_\tau \end{pmatrix} \tag{1.2}$$

The charged leptons (e^-, μ^-, τ^-) carry an electric charge of -1 and interact via electromagnetic and weak forces. Neutrinos $(\nu_e, \nu_\mu, \nu_\tau)$ are electrically neutral and interact exclusively via the weak force.
Both quarks and charged leptons, have their own anti-particles, which have the same mass as the matter components but opposite quantum numbers. Regarding neutrinos, it is still debated and to be proved whether they have a distinct anti-particle or they are Majorana fermions, being their own anti-particle.
The second class of particles is associated with mediatiors and mass. Spin 1 vector bosons, known as *gauge bosons*, are the mediators of three of the fundamental interactions, which arise naturally from the requirement of local gauge invariance. Among those fields we find the 8 massless gluons g mediating the strong interaction, the massless photon γ is associated to electromagnetic interaction and 3 massive vector bosons, $W^\pm$ and Z^0, with masses $m_W = 80.36$ GeV, $m_Z = 91.19$ GeV, mediating the weak interaction.
In conclusion, a spin 0 scalar particle, the *Higgs boson* H, with a mass $m_H = 125.13$ GeV, completes the particle spectrum, arising as a physical state from the

spontaneous breaking of electroweak symmetry.

The SM is a gauge theory, meaning that the Lagrangian is invariant under local transformations of the fields. The Lagrangian density describing the SM dynamics can be written as:

$$\mathcal{L}_{\text{SM}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - i\bar{\psi}\gamma_\mu D^\mu\psi + (D_\mu\phi)^\dagger(D^\mu\phi) - V(\phi) + \bar{\psi}_i y_{ij}\psi_j\phi + h.c., \quad (1.3)$$

The gauge group of the SM, under which such Lagrangian is invariant, is a non-abelian symmetry group defined as:

$$SU(3)_C \otimes SU(2)_L \otimes U(1)_Y \quad (1.4)$$

where $SU(3)_C$ describes the strong interaction via Quantum Chromodynamics (QCD) with subscript C standing for colour, and $SU(2)_L \otimes U(1)_Y$ represents the unified electroweak (EW) interaction, with L associated to the left component of fermions and Y to hypercharge.
The terms of the equation 1.3 above are presented in detail in the following sections.

1.1.1 Quantum Electrodynamics

The first and simplest gauge theory within the SM framework is Quantum Electrodynamics (QED). It describes the electromagnetic interactions between a charged massive fermion, with mass m and represented by a field $\psi(x)$, and the photon, the massless gauge boson represented by the vector field A_μ. Using the Lagrangian formalism [6], QED is constructed starting from the free Dirac Lagrangian for a massive fermion [7]:

$$\mathcal{L}_{\text{Dirac}} = \bar{\psi}(i\gamma^\mu\partial_\mu - m)\psi \quad (1.5)$$

In the equation above, $\bar{\psi} = \psi^\dagger\gamma^0$ is the Dirac adjoint of the field ψ and γ^μ are the Dirac matrices derived from Pauli matrices σ_i (i =1,2,3 and μ=0,1,2,3):

$$\gamma^0 = \begin{pmatrix} 0 & \mathbb{I} \\ \mathbb{I} & 0 \end{pmatrix}, \qquad \gamma^i = \begin{pmatrix} 0 & \sigma_i \\ -\sigma_i & 0 \end{pmatrix} \quad (1.6)$$

The Dirac Lagrangian 1.5 is invariant under a global $U(1)$ phase transformation $\psi(x) \rightarrow e^{ie\alpha}\psi(x)$, with e the electric charge of the fermion and α a constant phase. Such invariance is not preserved under a local $U(1)$ phase transformation $\psi(x) \rightarrow e^{ie\alpha(x)}\psi(x)$, where $\alpha(x)$ is a space-time dependent phase. In order to preserve the invariance, the partial derivative ∂_μ is replaced by the covariant derivative D_μ:

$$D_\mu = \partial_\mu + ieA_\mu(x) \quad (1.7)$$

where $A_\mu(x)$ is a vector gauge field representing the photon field, which transforms as $A_\mu(x) \rightarrow A_\mu(x) - \partial_\mu\alpha(x)$. Including the gauge-invariant kinetic term to take into account for the new field A_μ, we derive the full QED Lagrangian:

$$\mathcal{L}_{\text{QED}} = \bar{\psi}(i\gamma^\mu D_\mu - m)\psi - \frac{1}{4}F_{\mu\nu}F^{\mu\nu} \quad (1.8)$$

where $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ is the electromagnetic field strength tensor. The interaction term $-e\bar{\psi}\gamma^\mu\psi A_\mu$ rules the coupling of charged fermions to photons with a strength proportional to the electric charge e.
The invariance under a local $U(1)$ transformation gurantees, via the Noether's theorem [8], the existence of a conserved current and charge which, in the case of QED, are the electromagnetic current and the electric charge, respectively.

1.1.2 Quantum Chromodynamics

The building of the Lagrangian for QCD is similar to the one for QED: the Dirac Lagrangian is required to be invariant under a local symmetry group, which in this case is the non-Abelian $SU(3)_C$ group, where C represents the triplet color degree of freedom (red, green, blue), the conserved charge of QCD [9]. This theory models the strong interaction between *quarks*, fundamental constituents of hadrons represented as fermionic fields $q(x)$, and *gluons*, the massless gauge bosons represented as vector fields $G^a_{\mu\nu}(x)$ with $a = 1, ..., 8$. Similarly to QED, the local transformation of the quark field is defined as $q(x) \to q'(x) = e^{ig_s T^a \alpha_a(x)} q(x)$, with g_s the strong coupling constant and $\alpha_a(x)$ the local phase. The term T^a appearing in the previous transformation equation is associated to the generators of the $SU(3)_C$ group, which can be expressed via the Gell-Mann matrices as $T^a = \frac{1}{2}\lambda^a$; such generators satisfy $[T^a, T^a] = if^{abc}T^c$ with f^{abc} being the structure constants of $SU(3)$.
Also in this case, a covariant derivative is introduced to preserve the invariance of the Lagrangian under local transformations:

$$D_\mu = \partial_\mu + ig_s T_a G^a_\mu \tag{1.9}$$

The non-Abelian structure of the group, gives rise to the QCD field tensor:

$$F^a_{\mu\nu} = \partial_\mu G^a_\nu - \partial_\nu G^a_\mu - g_s f^{abc} G^b_\mu G^c_\nu \tag{1.10}$$

The presence of the non-linear structure factor f^{abc} both in the commutators and in the tensor, is associated to a particular feature of QCD: the triple and quadruple gluon self-interaction of gluons, a feature not present in QED.
Finally, the Lagrangian describing the dynamics of QCD is written as:

$$\mathcal{L}_{\text{QCD}} = \bar{q}\left(i\gamma^\mu D_\mu - m\right) q - \frac{1}{4} F^a_{\mu\nu} F^{a,\mu\nu} \tag{1.11}$$

The term g_s, cited before in the local transformation, is usually presented as $\alpha_s = \frac{g_s^2}{4\pi}$, which is the running coupling constant of QCD, whose value depends on the energy scale Q^2 of the process. The non-Abelian nature of QCD gives rise to a peculiar behavior of $\alpha_s(Q^2)$. In fact, at high energy scales (meaning short distances), $\alpha_s(Q^2)$ decreases logarithmically, giving rise to the *asymptotic freedom*, enabling accurate calculations via perturbative QCD (pQCD). On the other hand, at low energy scales (meaning long distances), $\alpha_s(Q^2)$ increases, giving rise to the *color confinement*, which prevents colored particles from existing as isolated free states. In this regime, quarks and gluons undergo fragmentation and hadronization, producing collimated sprays of energetic hadrons, known as *jets*.

1.1.3 Electroweak model

Making use of the Glashow-Weinberg-Salam (GWS) model [7, 10], it is possible to derive a unified description of *electromagnetic* interaction, already presented in Sec. 1.1.1, and the final third interaction, the *weak* one.
To derive the Electroweak (EW) Lagrangian, the steps to follow are similar to those already presented for QED and QCD. In the first place, the symmetry group under which the Lagrangian is invariant is defined, which is $SU(2)_L \otimes U(1)_Y$ in this case, with Y known as hypercharge, the quantum number of EW interactions.
Secondly, the description of fermions and gauge bosons is defined. Concerning the latter, we have four bosons: W_μ^i with $i = 1, 2, 3$ associated to $SU(2)_L$ and B_μ associated to $U(1)_Y$, with coupling constants g and g' respectively. Moving to fermions, they are chirality eigenstates. This means that they can be expressed as left-handed and right-handed components, which are defined as $\psi_{L,R} = \frac{1}{2}(1 \mp \gamma^5)\psi$, with $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$. The left-handed components of fermions transform as doublets under $SU(2)_L$ and are described as $L = (\nu, l^-)$ for leptons and $Q = (u, d)$ for quarks. They have quantum numbers $(T = 1/2, T_3 = \pm 1/2)$, with T being the isospin. The right-handed components, instead, transform as singlets under $U(1)_Y$, with quantum numbers $(T = 0, T_3 = 0)$.
The transformation of the fermion components is defined in a compact way as:

$$\begin{aligned} \psi_L(x) &\longrightarrow \psi'_L(x) = e^{i\vec{\alpha}(x)\cdot\vec{\tau} + i\beta(x)Y}\psi_L(x), \\ \psi_R(x) &\longrightarrow \psi'_R(x) = e^{i\beta(x)Y}\psi_R(x) \end{aligned} \tag{1.12}$$

where $\vec{\tau} = \frac{1}{2}\vec{\sigma}$ are the weak isospin, generators of $SU(2)_L$.
Electric charge Q, weak isospin T_3, and weak hypercharge Y are connected by the relation:

$$Q = T_3 + \frac{Y}{2} \tag{1.13}$$

The following step towards the EW Lagrangian, is to introduce the covariant derivative to preserve the invariance of the Lagrangian under local transformations. The derivative is defined as:

$$D_\mu = \partial_\mu + ig\frac{\tau^a}{2}W_\mu^a(x) + ig'\frac{Y}{2}B_\mu(x) \tag{1.14}$$

At this point, the EW Lagrangian can be written as:

$$\mathcal{L}_{\text{EW}} = -\frac{1}{4}W^{a,\mu\nu}W^a_{\mu\nu} - \frac{1}{4}B^{\mu\nu}B_{\mu\nu} + i\bar{\psi}\gamma^\mu D_\mu\psi \tag{1.15}$$

where $W^{a,\mu\nu}$ and $B^{\mu\nu}$ are the field strength tensors associated to the gauge bosons W_μ^a and B_μ, respectively.
In conclusion, if a rotaion is applied to the W_μ^a and B_μ fields, the four physical gauge bosons $\gamma, Z, W^\pm$ are obtained as:

$$\begin{pmatrix} A_\mu \\ Z_\mu \end{pmatrix} = \begin{pmatrix} \cos\theta_W & \sin\theta_W \\ -\sin\theta_W & \cos\theta_W \end{pmatrix}\begin{pmatrix} B_\mu \\ W_\mu^3 \end{pmatrix}, \quad W_\mu^\pm = \frac{1}{\sqrt{2}}(W_\mu^1 \mp iW_\mu^2) \tag{1.16}$$

The parameter appearing in the previous equation is the Weinberg angle θ_W, which is defined as $\tan\theta_W = g'/g$.

The EW Lagrangian 1.15 does not provide a proper description of the mass of the bosons involved, which is in contrast with the experimental observations of massive $W^{\pm}$ and Z^0 bosons. The solution to this issue is provided by the Brout-Englert-Higgs mechanism (BEH), which is presented in the following section.

1.1.4 The Brout-Englert-Higgs mechanism

The BEH mechanism provides a theoretical framework to generate masses for the gauge bosons and fermions of the SM, while preserving the local gauge invariance of the Lagrangian. The mechanism is based on the concept of spontaneous symmetry breaking (SSB) of the group $SU(3)_C \otimes SU(2)_L \otimes U(1)_Y$ into $SU(2)_L \otimes U(1)_{EM}$ [11]. The SSB is realized by introducing a complex scalar field ϕ, which is a doublet under $SU(2)_L$ and carries hypercharge $Y = 1/2$. It is defined as:

$$\phi = \begin{pmatrix} \phi^+ \\ \phi^0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} \phi_1 + i\phi_2 \\ \phi_3 + i\phi_4 \end{pmatrix} \tag{1.17}$$

The potential associated to the scalar field is defined as:

$$V(\phi) = -\mu^2 \phi^\dagger \phi + \frac{\lambda}{2} \left(\phi^\dagger \phi\right)^2 \tag{1.18}$$

where μ^2 and λ are constants. Among the two constants, λ must be positive to ensure the stability of the potential, while μ^2 can be either positive or negative. The shape of the potential is determined by the sign of μ^2. If $-\mu^2 > 0$, the potential has a single minimum at $\phi = 0$, preserving the symmetry. If $-\mu^2 < 0$, the potential gets the well-known mexican hat shape presented in Fig 1.2.

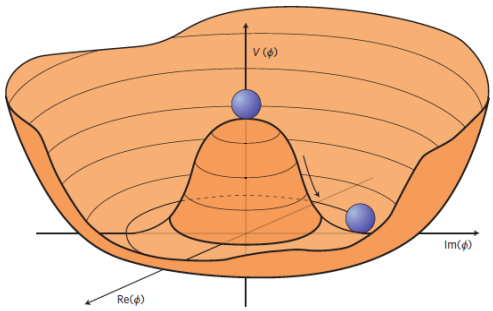


Figure 1.2 Representation of the Higgs potential $V(\phi)$. Figure from Ref. [12].

In the latter case, the vacuum state, as known as the minimum of the potential, is degenerate and associated to a non-zero value of the scalar field ϕ. In particular, the vacuum expectation value (VEV) v of the Higgs field, defined as $v = \sqrt{\mu^2/\lambda}$, is not null. As a direct consqeuence, the global $U(1)_Y$ symmetry is spontaneously broken. The VEV is given by:

$$\phi^\dagger \phi = \frac{\mu^2}{2\lambda} = \frac{v^2}{2}, \quad \phi_0 = \frac{v}{\sqrt{2}} \tag{1.19}$$

Since the theory is built to be gauge invariant, in order to remove the unphysical degrees of freedom, the unitary gauge transformation is applied to the scalar field ϕ.

In this gauge, the Higgs field can be expressed as:

$$\phi = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v + h(x) \end{pmatrix} \tag{1.20}$$

where $h(x)$ is the Higgs boson scalar field.
Using the definition of ϕ presented in Eq. 1.20, the EW Lagrangian density can be written as:

$$\begin{aligned} &\mathcal{L}_{\text{SM}} = -\tfrac{1}{4} W^{a,\mu\nu} W^{a}_{\mu\nu} - \tfrac{1}{4} B^{\mu\nu} B_{\mu\nu} + i\overline{\psi}\gamma^{\mu} D_{\mu}\psi + \\ &-Y^{l}_{ij}\overline{L_i}\phi\, e_j - Y^{\nu}_{ij}\overline{e_i}\phi\, L_j - Y^{d}_{ij}\overline{Q_i}\phi\, d'_j - Y^{u}_{ij}\overline{Q_i}\phi\, u_j + \text{h.c.} \\ &+(D^{\mu}\phi)^{\dagger}(D_{\mu}\phi) - V(\phi) \end{aligned} \tag{1.21}$$

where the term D_μ is the covariant derivative defined in Eq. 1.14 and Y_{ij} are the Yukawa couplings. Each line of the Lagrangian density 1.21 has as specific role. The first line represents the gauge bosons and how they interact with fermions, the second one contains the Yukawa couplings which is associated to Higgs coupling with fermions, and the third line contains the kinetic term for the Higgs field and its potential.
The Higgs potential $V(\phi)$, defined in Eq. 1.18, with che chosen gauge, can be expressed as:

$$V(h) = \frac{1}{2} m_h^2 h^2 + \frac{m_h^2}{2v} h^3 + \frac{m_h^2}{8v^2} h^4 \tag{1.22}$$

The m_h term appearing in the previous equation is the mass of the Higgs boson, which is a free parameter of the SM, depending on the parameters of the potential, as $m_h = \sqrt{2\lambda v^2} = \sqrt{2\mu^2}$. The cubic and quartic terms in h represent self-interactions of the Higgs boson.
The Higgs boson mass has been measured by the ATLAS and CMS collaborations [1, 2] to be $m_h = (125.13 \pm 0.11)$ GeV, which is consistent with the SM predictions. From this measurement, the value of VEV v can be derived as $v \approx 246$ GeV.

1.2 Hints of Physics Beyond the Standard Model

The SM presented in the previous section 1.1 is a very successful theory, which has been confirmed by experimental observations with a very high precision. Despite its success, the SM probably is not a complete theory, given the many open questions that are not answered by the SM and motivate the search for Physics Beyond the Standard Model (BSM).
Among these questions, we find the nature of the Higgs boson, whether it is the only scalar particle or it is part of an extended Higgs sector. In fact, many extensions of the SM predict the existence of additional scalar or pseudo-scalar states, such as two-Higgs-doublet models (2HDMs), which are currently studied at the LHC by the ATLAS collaboration [13].
Together with theories to be proven, as the one just cited, there are many other proven by astrophysical and cosmological observations. The first one is the existence of Dark Matter (DM), which is known to compose 26.4% of the Universe, together

with a 68.7% of Dark Energy and a 4.9% of ordinary matter [14]. The presence of DM is inferred from its gravitational effects on visible matter, such as galaxies rotation curves [15], galaxy-galaxy gravitational lensing [16] and the Bullet Cluster [17], the result of the collision of two galaxy clusters. In order to observe DM at collider experiments, the approach, known as indirect search, is to detect decay products of DM particles. This type of search was not able to provide any evidence of DM particles yet, leaving it an ongoing study at the LHC.
Another open question is the matter-antimatter asymmetry in the Universe, also known as baryon asymmetry. In fact, in the early Universe there was balance between matter and antimatter, but today the Universe, after the cooling down of the Universe, the balance is slightly broken, in favour of matter. Such unbalance could be associated to a violation of the CP symmetry, which is a combination of charge conjugation C and parity P. This motivates the search for new sources of CP violation, which could be provided by BSM theories.
One final question is the hierarchy problem, which is associated to the Higgs boson mass. In fact, the Higgs boson mass value is subject to quantum loop corrections, which are quadratically divergent. This means that the Higgs boson mass receives contributions from all energies up to the Planck scale (10^{19} GeV). Since the Higgs boson mass has been measured to be 125 GeV, an unnatural fine-tuning of the parameters of the SM would be required with large cancellations between different terms in radiative correction. This motivates the search for new physics at the TeV scale, which could provide a natural explanation for the Higgs boson mass.

1.2.1 New Physics at colliders

After reading the BSM cases presented above, and many others not cited, it appears to be quite clear that New Physics Beyond the Standard Model should exist and, in principle, it should be accessible at the energy scales probed by the LHC. Given the lack of observations at CERN experiments, it is worth analysing the approach used to probe NP. This kind of experiments are conducted with two main different approaches: *precision measurements*, which are sensitive to small deviations from the SM predictions, and *direct searches*, which aim to produce new particles in high-energy collisions and probe them. The latter direction has been traditionally pursued by targeting a specific BSM model, which is known as *model-dependent* search. In this case, the search is optimized for a specific model and its parameters. The results of such searches are usually presented as exclusion limits on the space of parameters of the model.
A suitable alternative to model-dependent searches is the *model-independent* or *model-agnostic* approach, which searches for deviations from the SM predictions in a general way, without assuming any specific BSM model. This approach is more flexible and can be sensitive to a wider range of BSM scenarios. This type of search is gaining more and more interest in recent years in the LHC program for Run 3.
A model-agnostic search for NP is the topic of this thesis, with the goal to prove its flexibility and sensitivity. Two possible BSM theories have been chosen as benchmark models to test the strategy, Heavy Vector Triplets (HVT) which could explain new narrow resonances, and Dark Jets (DJ) which are QCD-like jets produced by dark

sectors.

1.2.2 Heavy Vector Triplets

Popular studied BSM scenarios are those presented in Simplified Models [18], which assume that NP are associated with new narrow resonances. Such assumption has the direct consequence of reducing the number of free parameters of the model, making it more predictive and easier to test.
The simplest and most studied Simplified Model is the Heavy Vector Triplet (HVT) model [19], which predicts the existence of electroweakly charged spin 1 bosons with a behavior similar to the SM W and Z bosons. This model introduces in the SM Lagrangian the presence of the real vector field V_μ^a $(a = 1, 2, 3)$, which is a triplet under $SU(2)_W$ and has zero hypercharge. Making a combination of these components, the physical states of the HVT model are obtained as:

$$V_\mu^\pm = \frac{V_\mu^1 \mp iV_\mu^2}{\sqrt{2}}, \quad V_\mu^0 = V_\mu^3 \tag{1.23}$$

The HVT undergo Drell-Yan decays with highest branching ratios with two leptons, two quarks or two bosons in the final state.
ATLAS and CMS collaborations have conducted many analyses to search for HVT decaying mainly in fully hadronic final states, which are the most abundant ones. The results of these searches are usually presented as exclusion limits at 95% confidence level on the mass of the HVT bosons, which are currently excluded in the range $[1.3, 3.1]$ TeV by ATLAS [20] and $[3.3, 3.5]$ TeV by CMS [21]. A typical plot of these kind of searches can be found in Fig. 1.3, showing the result of the ATLAS analysis.

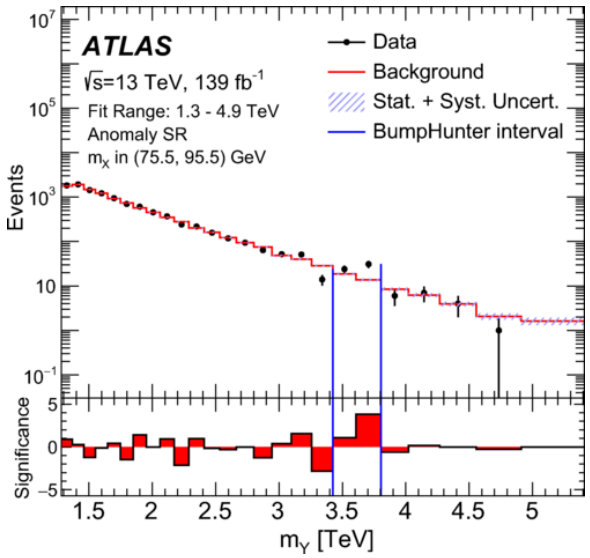


Figure 1.3 BumpHunter scan in the Y boson invariant mass distribution within the HVT framework. Figure from Ref. [20].

1.2.3 Dark Jets

The dark QCD model, a strongly coupled dark sector, has gained interest in the last years as a possible explanation for DM. [22] In this model, the dark sector is composed of dark quarks and gluons, which are charged under a new gauge group $SU(N)_D$ and interact via a new strong force, similar to QCD.
The dark quarks can be pair-produced at colliders via a mediator particle. Once produced, they undergo a shower and hadronization process, similar to QCD, producing collimated sprays of dark hadrons, known as dark jets. This process is predicted to happen at a scale Λ_D, which is the confinement scale of the dark sector, with the assumption that $\Lambda_D << \sqrt{s}$, where $\sqrt{s}$ is the center-of-mass energy of the collision.
The commonly known portal is the mediator particle involved in the interaction between the SM and the dark sector. The experimental signature of those processes at colliders will be jets with missing energy or general features not compatible with QCD.
ATLAS and CMS collaborations have conducted many analyses to search for dark jets. In particular, ATLAS has conducted a search for a $Z' \rightarrow q_D\overline{q}_D$ resonance [23], where q_D is a dark quark and Z' is probed in the range of $[1.5, 3.5]$ TeV. The observed limit on the mass for such new resonance has been set up to 3.5 TeV, as presented in Fig. 1.4.

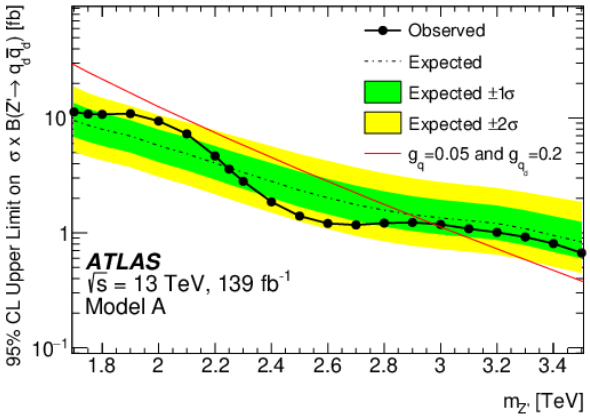


Figure 1.4 Upper limits at 95% CL on the cross section times the branching ratio for a dark QCD process. Figure from Ref. [20].

Chapter 2

The LHC and the ATLAS experiments

The history of the European Organization for Nuclear Research (CERN) traces back to the post-WWII era, when European physicists, including Edoardo Amaldi, Niels Bohr and Pierre Auger, promoted a collaborative international laboratory dedicated to fundamental science. CERN was established in September 1954 and today it hosts the world's largest and most energetic accelerator complex, centered around the Large Hadron Collider (LHC).
In the following, after a description of the LHC machine, the discussion covers the architecture of the ATLAS (A Toroidal LHC ApparatuS) detector, one of the four major experiments at CERN, including its main sub-systems and simulation procedures. A special focus is given to the proton-proton collision dataset collected by ATLAS at a center-of-mass energy of 13.6 TeV during LHC Run 3, which forms the basis for this study.

2.1 The LHC

The Large Hadron Collider (LHC) at CERN, with its 27 kilometres ring [25], is the world's largest particle accelerator. It has four major interaction points which host the major CERN experiments: ATLAS, CMS, LHCb and ALICE. LHC is situated about 100 metres underground for economic[1]and geological reasons, as well as to provide radiation shielding. With its first run on 10 September 2008, it currently accelerates hadrons. Depending on the run, protons or lead nuclei are collided, with a center-of-mass energy of 13.6 TeV and 5.36 TeV respectively. In order to reach such high energies, a complex system of pre-acceleration is adopted, as presented in Fig. 2.1.

[1]Building LHC on the surface would have required the purchase of properties to do so and an under the open sky structure with a 27 km circumference would have been more expensive than agreeing with landowner and excavate a tunnel underground.

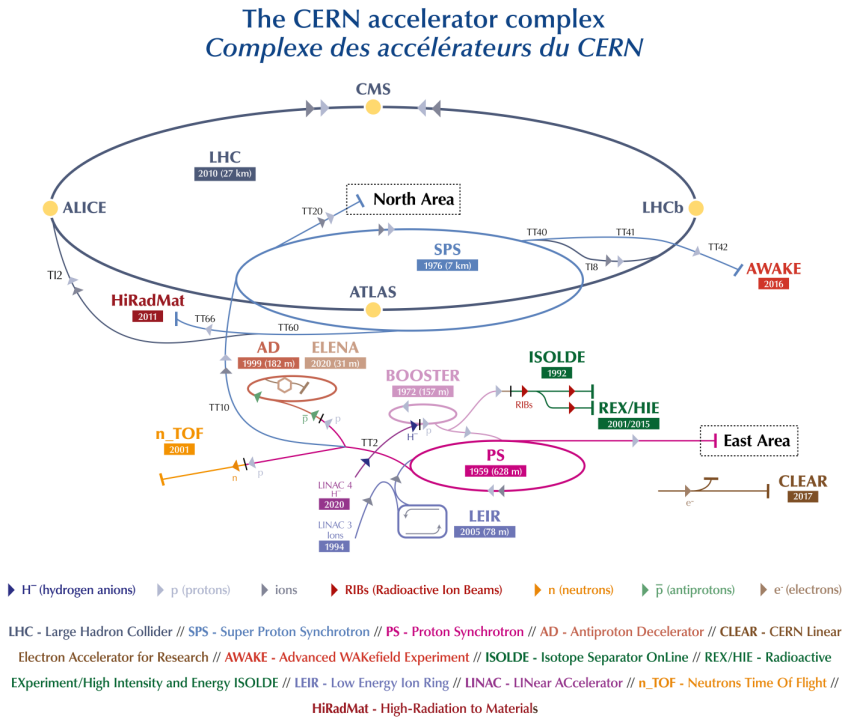


Figure 2.1 CERN accelerating system. Figure from Ref. [24].

2.1.1 CERN's accelerator system

For proton-proton (pp) collisions, the process begins with the extraction of negative hydrogen ions (H^-), which are initially accelerated to an energy of 160 MeV by the Linear Accelerator 4 (LINAC4). Upon injection into the Proton Synchrotron Booster (PSB), the ions are stripped of their two electrons leaving only the bare protons, which are grouped into bunches and boosted to an energy of 2 GeV. The protons then enter the Proton Synchrotron (PS), reaching an energy of 25 GeV. The final step takes place in the Super Proton Synchrotron (SPS) that accelerates the proton bunches up to the LHC injection energy of 450 GeV. Finally, the proton bunches are injected into the two counter-rotating beam pipes of the LHC, where they are accelerated to their maximum operational energy of 6.8 TeV per beam.
In heavy-ion operations (primarily based on lead nuclei, ^{208}Pb), ions are initially accelerated by the Linear Accelerator 3 (LINAC3) to an energy of 4.2 MeV per nucleon. The nucleons are then accumulated and boosted to 72 MeV per nucleon inside the Low Energy Ion Ring (LEIR). In the next stage, the PS accelerates ions to 5.9 GeV per nucleon, before transferring them to the SPS where they reach an energy of 177 GeV per nucleon. When injected into the main LHC ring, the heavy-ion beams are accelerated to a final energy of 2.68 TeV per nucleon.
Particles are accelerated up to such high energies by means of radio-frequency (RF) cavities, resonant structures generating an oscillating electric field, with a fixed frequency of 400MHz, which accelerates charged particles each time they pass through. Particles with the target energy passing trough the cavity while the field is zero,

will not be accelerated, while more (less) energetic particles will be decelerated (accelerated). Using this technique, it is ensured that particles stays in the cited bunches.

2.1.2 The LHC parameters

The LHC is composed of two circular beam pipes, denominated Beam 1 and Beam 2. Inside them, two particle beams, composed of a maximum of 2808 bunches with approximately 1.15×10^{11} protons per bunch and spaced by 25 ns, travel separately close to the speed of light in opposite directions. There are eight Interaction Points (IPs) along the LHC ring, but only four of them are used for collisions, hosting the main LHC experiments: ATLAS at IP1, ALICE (A Large Ion Collider Experiment) at IP2, CMS (Compact Muon Solenoid) at IP5 and LHCb (LHC beauty) at IP8. The beam pipes and the interaction points are presented in Fig. 2.2.

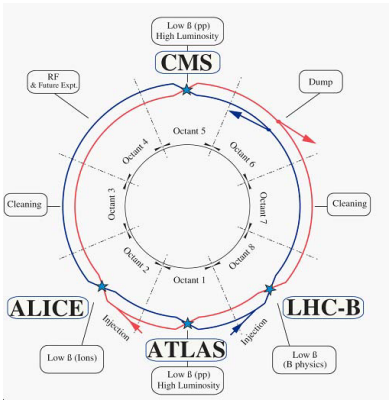


Figure 2.2 Schematic layout of the LHC (Beam 1 clockwise, Beam 2 anticlockwise). Figure from Ref. [26].

Due to the very high speed, when the two beams collide, the energy of the center of mass, known as $\sqrt{s}$, is twice the energy of each beam (which is the same). In fact, knowing that $E^2 = m^2c^4 + p^2c^2$ and using the natural unit system ([m]=eV/c^2, [p]=eV/c), we get: $E^2 = m^2 + p^2 \simeq p^2$, where at the end an ultra-relativistic particle (p»m) was assumed. This information is used in the norm of the four-momentum of two colliding particles:

$$\sqrt{s} = \sqrt{P_\mu P^\mu} = \sqrt{E_{tot}^2 - |\vec{p_{tot}}|^2} = \sqrt{(E_1 + E_2)^2 - |\vec{p_1} - \vec{p_2}|^2} \simeq 2E \qquad (2.1)$$

where we assumed $E_1 = E_2$. The 6.8 TeV of energy reached by LHC protons confirm the assumption p»m and it leads to the known value of $\sqrt{s} = 13.6$ TeV.
Accelerated beams are directed and curved by superconducting electromagnets with a magnetic field of up to 8.3 T [27]: 1232 dipole magnets and 392 quadrupole magnets are used to maintain the beams focused. Magnets are kept at a temperature of 1.9 K to ensure a 11600 A electric current which is needed for the intense magnetic field. Despite their essential contribution, magnets also represent one of the main limitations of this apparatus. In fact, from the Lorentz force equation $\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$, it is possible to get a connection between the external magnetic

field $\vec{B}$ (the one generated by superconducting magnets), the momentum p of the particle circulating inside the beams and its radius of curvature R, which is:

$$p[GeV] = 0.3B[T]R[m] \tag{2.2}$$

It is evident from Eq. 2.2 that, to keep a constant momentum p of the particles circulating inside the beams, a higher magnetic field B is required in case of reduction of the radius of curvature R. Unfortunately, current technologies set a limit to the magnetic field generated by a magnet, limiting also the minum dimension of an accelerator ring.
Performance of the LHC is evaluated based on an essential parameter: the instantaneous luminosity $\mathcal{L}$. It defines the number of collisions per cm^2 and per second produced in the collider and thus it is measured in $cm^{-2}s^{-1}$. It is computed as follows [28]:

$$\mathcal{L} = \frac{N_b^2 n_b f_{rev} \gamma}{4\pi\epsilon_n \beta^*} F \tag{2.3}$$

where: N_b is the number of particles in a bunch, n_b is the number of bunches in a beam, ϵ_n is the normalized transverse beam emittance, β^* is the beta function at the collision point, γ is the Lorentz relativistic factor computed as $\gamma = 1/\sqrt{1 - v^2/c^2}$, f_{rev} is the revolution frequency. There is a connection between the beta function and the transverse beam size σ, which is $\sigma = \sqrt{\epsilon_n \beta^*/\gamma}$. F represents the geometric luminosity reduction factor:

$$F = \frac{1}{\sqrt{1 + \left(\frac{\theta_c \sigma_z}{2\sigma^*}\right)^2}} \tag{2.4}$$

where θ_c is the full crossing angle at the interaction point, σ_z is the RMS bunch length and σ^* is the transverse RMS beam size at the interaction point. If during operations the instantaneous luminosity goes below a fixed threshold, beams are dumped and a new fill of the LHC is performed.
During LHC Run 3, $\mathcal{L}$ gets the value of 10^{34} cm$^{-2}s^{-1}$ for both ATLAS and CMS experiments, while for LHCb and ALICE it is 10^{32} cm$^{-2}s^{-1}$ and 10^{27} cm$^{-2}s^{-1}$ respectively.
The instantaneous luminosity is linked with the number of collisions per second :

$$\frac{dN}{dt} = \mathcal{L}\sigma_{tot} \tag{2.5}$$

where σ_{tot} is the total cross section, which is the measure of the probability of a specific interaction process to happen. If the instantaneous luminosity is integrated over time, the number of events collected is obtained:

$$N_{events} = \sigma_{tot} \int \mathcal{L} dt \tag{2.6}$$

2.1.3 LHC Run Schedule

The LHC operations are structured into extended physics data-taking campaigns named *Runs*, which are separated by planned technical maintenance and machine

upgrade periods, which are known as Long Shutdowns (LS). It was Run 1, from 2010 to 2013, that started the LHC physics program, operating at center-of-mass energies of $\sqrt{s} = 7$ TeV and 8 TeV, with a delivered integrated luminosity of 28 fb^{-1}. Run 1 culminated in the discovery of the Higgs boson by the ATLAS and CMS collaborations. Following upgrades during LS1, meant to enable higher-energy operations, Run 2 took place between 2015 and 2018 at $\sqrt{s} = 13$ TeV, delivering an integrated luminosity of approximately 156 fb^{-1}. After another round of upgrades during LS2 the collider entered the Run 3 era in 2022, which lasted until June 2026, operating at an increased center-of-mass energy of $\sqrt{s} = 13.6$ TeV. After having delivered an integrated luminosity of approximately 351 fb^{-1}, the LHC was turned off for the beginning of LS3. This new stop will be used to prepare the LHC for the High-Luminosity LHC (HL-LHC) era, which is expected to start in 2029 and will last until 2042. The HL-LHC will operate at the same center-of-mass energy of Run 3 and is expected to deliver an integrated luminosity of 3000 fb^{-1} during the future Runs.
A graph showing the LHC delivered integrated luminosity during the different Runs, together with the recorded luminosity by ATLAS, is shown in Fig. 2.3.

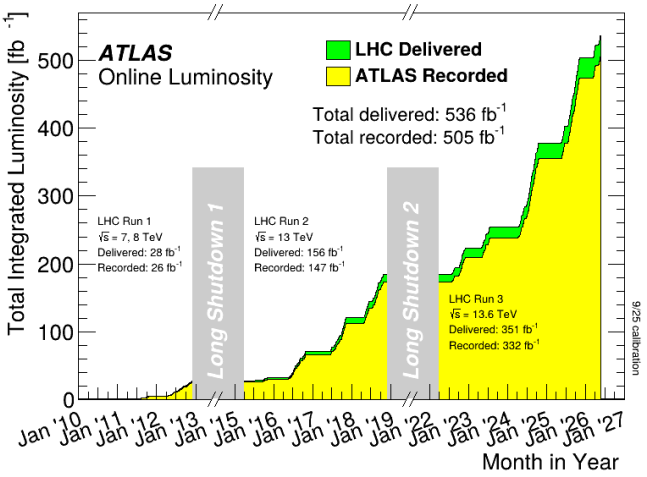


Figure 2.3 Cumulative integrated luminosity delivered to and recorded by the ATLAS Experiment during LHC Run 1, 2 and 3. Figure from Ref. [29].

2.2 *pp* collisions at the LHC

As presented in Sec. 2.1.2, at the LHC protons are collided every 25 ns at a center-of-mass energy of $\sqrt{s} = 13.6$ TeV. Theoretical calculations to model such events using QCD are not trivial, since protons are not elementary particles but composite states of quarks and gluons, collectively referred as *partons* [30]. The canonical approach used with other theories, the *perturbation theory*, can still be used in the case of high-energy interactions, known as *hard scattering*, where the strong coupling constant α_s is small enough to allow a perturbative expansion. It fails at low energy scales, in the so called *soft scattering*, where α_s becomes large and the perturbative expansion diverges.

In these cases, the *factorization theorem* allows to compute the total cross-section of *pp* collisions as a convolution of short-distance partonic matrix elements with long-distance Parton Distribution Functions (PDFs), which describe the probability of finding a parton with a given momentum fraction x inside the proton.
The situation described above gets a degree of complexity when considering that, in a single bunch crossing, multiple *pp* collisions can occur. This phenomenon, known as *pile-up*, is discussed in Sec. 2.2.1.
As said, the calculation of the total cross-section of *pp* collisions is not trivial. For this reason, event generators are used to simulate the physics of *pp* collisions, as described in Sec. 2.3.8 for ATLAS.
In Fig. 2.4 a Feynman diagram of a *pp* collision is shown, where the two protons are represented as two bunches of partons, among which only two of them collide and produce the final state.

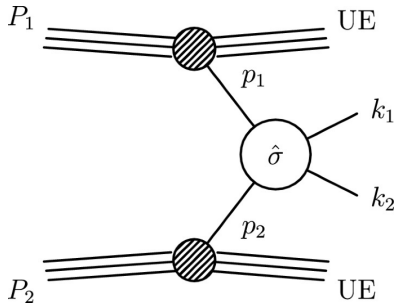


Figure 2.4 Feynman diagram of a *pp* collision. Figure from Ref. [31].

2.2.1 Pile-Up and hadronisation

The partons coming out of the hard scattering process emit gluon radiation and produce quark pairs in a cascading parton shower (Sec. 2.2.3).
When approaching energies compatible with the QCD scale $\Lambda_{\text{QCD}} \approx 200/300$ MeV, the soft scattering regime is reached, with the divergence of the perturbative expansion and the araising of colour confinment. For this reason, colour charged partons must bind into colour-neutral hadrons. This non-perturbative transition is known as *hadronisation* [31]. The processes and collisions not related to the primary hard scatter are defined as *underlying events*, which can rise from multiple sources, such as initial of final state radiation. This type of events, together with soft scattering processes from other partons, are responsible for the Pile-Up (PU) phenomenon. PU degrades event reconstruction and it has to be carefully controlled to ensure a correct physics analysis. When referring to PU, the average number of *pp* collisions per bunch crossing is presented, using the symbol $\langle\mu\rangle$.
In Fig. 2.5, a plot showing the PU distribution in ATLAS from Run 1 to Run 3 is presented. It is evident that the average number of *pp* collisions per bunch crossing has increased over the years, reaching a value of $\langle\mu\rangle \approx 62$ during Run 3.

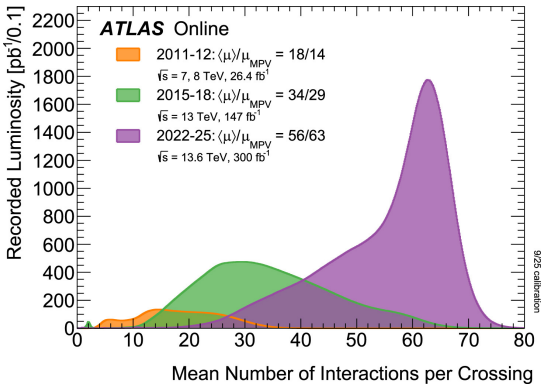


Figure 2.5 Pile-Up distribution in ATLAS from Run 1 to Run 3. Figure from Ref. [32].

2.2.2 Electromagnetic Showers

Before discussing the hadronic showers, which are essential in the formation of jets, the main focus of this thesis, it is important to introduce the electromagnetic version of them. Electromagnetic showers develop when high-energy electrons or photons interact with heavy materials. They are the basis of the electromagnetic calorimeter design, which is used to measure the energy of electrons and photons (Sec. 2.3.5). The passive material, or absorber, is typically a high-Z material with a high probability of interacting with the incoming particles. These interactions reduce particles' energy and slowly stop them.
Particles involved in the showering process can undergo two main processes: the first one is bremsstrahlung and the other is pair production. Electrons (and positrons) lose energy via bremsstrahlung, which can be defined as the emission of photons (radiation). Those photons undergo pair production, converting them into electron-positron pairs which then emit more photons via bremsstrahlung. This cycle repeats until electrons and photons loose their energies.
The count of particles involved in the cascade grows exponentially, forming a literal *shower* of particles, as shown in Fig 2.6.

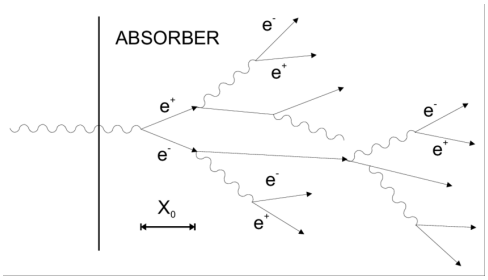


Figure 2.6 Schematic of an electromagnetic shower starting from a photon.

The fundamental parameter of the electromagnetic shower is the radiation length X_0, the mean distance over which an electron loses a factor $1/e$ of its energy. A typical EM shower has a longitudinal profile of $\sim 20\ X_0$. The calorimeter must be deep enough to contain the entire shower and measure the energy of the incoming particle.

2.2.3 Hadronic Showers

Hadronic showers develop when particles sensible to the strong interaction collide with nuclei in the calorimeter passive absorber. These inelastic collisions initiate a nuclear cascade with complex modelling. The shower, in fact, contains both an electromagnetic and a hadronic component. Neutral pions (π^0), for instance, decay immediately into photon pairs, initiating the electromagnetic sub-showers presented before. Charged pions ($\pi^{\pm}$), instead, continue to undergo strong interactions, producing secondary hadrons.
The characteristic scale of the hadronic shower development is the interaction length λ, a scale much larger than X_0. Again, the hadronic calorimeter needs to be deep enough to manage the containment of the entire shower.

2.3 The ATLAS detector

A Toroidal LHC ApparatuS (ATLAS) [33] is the world's largest particle detector, situated at Interaction Point 1 at CERN. 44m in length and 25m in height are required to investigate collisions between particles accelerated by the LHC, to study phenomena such dark matter, the Higgs boson and top quark. The ATLAS detector is composed of multiple sub-detectors organized in layers with a cylindrical symmetry around the beam pipe. Among the main sub-detectors we can find: the Inner Detector (ID), the Electromagnetic Calorimeter (ECal), the Hadronic Calorimeter (HCal) and the Muon Spectrometer (MS), as shown in Fig. 2.7.

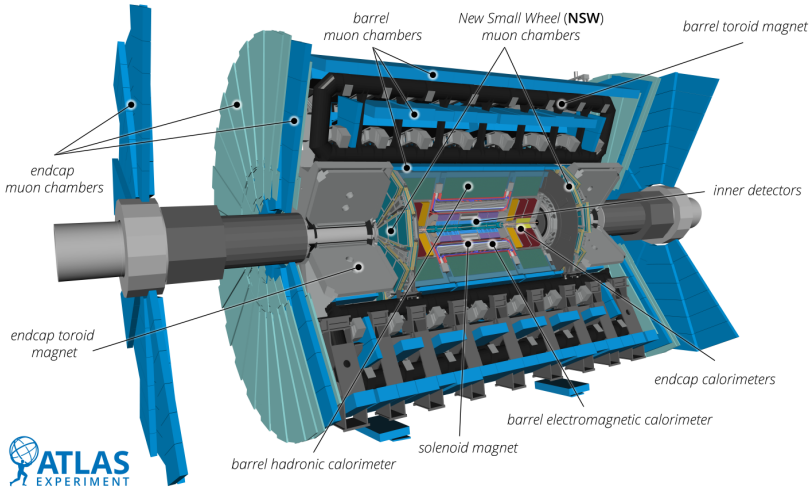


Figure 2.7 ATLAS schematic. Figure form Ref. [34].

2.3.1 Coordinate system

Before analysing the ATLAS detector, it is important to introduce the coordinate system used to describe the position of particles and their momenta inside the sub-detectors. In Fig. 2.8 an overview of the system is presented.

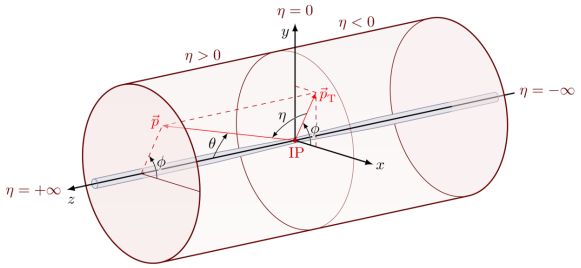


Figure 2.8 Schematic of ATLAS coordinate system

Cartesian axes (*x,y,z*) are constructed as follows. The origin is set in the interaction point (IP). From it, the z-axis departs parallel to the beam tunnel with positive z-values in anticlockwise direction. As a direct consequence, the x-y plane is perpendicular to it, with positive x-values towards the center of the LHC and positive y-values upward. On this Cartesian system, a more natural cylindrical coordinate system (r, θ, ϕ) is defined to describe geometric position and components of particle momenta. r is the absolute distance from the interaction point. ϕ is the azimuth angle, measured around the z-axis, with values $[-\pi, \pi]$. θ is the polar angle, measured starting from the z-axis, with values $[0, \pi]$.
Two more coordinates are defined to describe the position of particles in the detector: the rapidity y and the pseudo-rapidity η. They are defined as:

$$y = \frac{1}{2} ln\left(\frac{E + p_z}{E - p_z}\right) = \text{arctanh}\left(\frac{p_z}{E}\right) \tag{2.7}$$

$$\eta = -ln\left[tan\left(\frac{\theta}{2}\right)\right] = \text{arctanh}\left(\frac{p_z}{|\,\vec{p}\,|}\right) \tag{2.8}$$

where $\vec{p}$ is the three-momentum of the particle and p_z is its component along the z-axis. E represents the energy of the particle. y and η get values in the range $[-\infty, \infty]$, being zero for $\theta = \pi/2$ and $\pm\infty$ for $\theta = 0$ or $\theta = \pi$.
A schematic of the connection between the pseudo-rapidity and the polar angle θ is shown in Fig. 2.9.

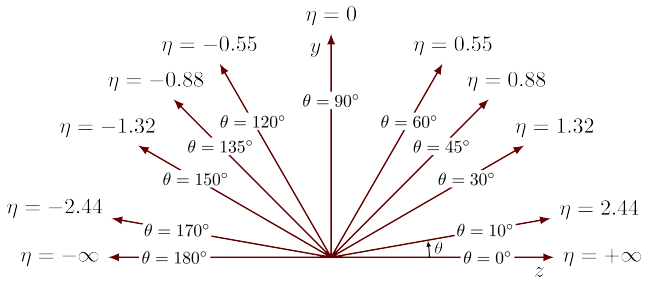


Figure 2.9 Schematic of pseudo-rapidity as a function of the polar angle θ.

These latter two variables are introduced because the difference of rapidities between two particles is a Lorentz invariant quantity under boosts along the z-axis. Since

Lorentz boosts are applied to move from the laboratory frame to the center-of-mass frame of the colliding particles, η and y are widely used instead of the polar angle θ. In case of an ultra-relativistic particle, the rapidity and the pseudo-rapidity are approximately equal, $\eta \approx y$, as shown in Eq. 2.7 and Eq. 2.8. The main difference between the two is that the rapidity depends both on the energy of the particle and its momentum, a combined measurement not trivial at high energies, while the pseudo-rapidity depends only on the particle's momentum. For these reasons, the pseudo-rapidity is preferred to the rapidity, making (r, η, ϕ) the used coordinate system. With such choice of coordinates, the detector components placed at high values of pseudo-rapidity are addressed as *forward region*.
The pseudo-rapidity and the azimuth angle, are used to define a geometrical distance, applied when computing the separation between two physics objects:

$$\Delta R = \sqrt{\Delta\eta^2 + \Delta\phi^2} \tag{2.9}$$

Prior to an analysis of materials, some more useful quantities are defined. The first one is the transverse momentum p_T, which is the projection of the momentum of the particle in the transverse plane (x, y) of the interaction, defined as:

$$p_T = \sqrt{p_x^2 + p_y^2} \tag{2.10}$$

The importance of this variables relies in the fact that at the LHC collisions happen mostly along the z-axis, generating Lorentz boosts mainly along this axis. This means that the transverse plane is not affected by these boosts, making p_T a Lorentz invariant quantity.
Similarly, the transverse energy E_T is defined as the projection of the energy of the particle in the transverse plane (x, y) of the interaction, defined as:

$$E_T = E \cdot \frac{p_T}{|\vec{p}|} \tag{2.11}$$

A fundamental variable for the purposes of this thesis is the invariant mass m_{inv}, which is associated to the energy of a system of particles. In the case of two particles, using their four-momenta P_1 and P_2, m_{inv} is defined as:

$$m_{inv} = \sqrt{(P_1 + P_2)^2} = \sqrt{(E_1 + E_2)^2 - |\vec{p_1} + \vec{p_2}|^2} \tag{2.12}$$

The invariant mass is a Lorentz invariant quantity, which is used to measure the mass of a resonant state, composed of a particle decaying into two others. In fact, the invariant mass of the system will be equal to the mass of the mother particle.
Based on what has been discussed on Lorentz boosts at the LHC, the sum of the the transverse momenta of the events is expected to be zero, since the initial state of the collision has no transverse momentum. For this reason, if a particle is not detected, as it happens with neutrinos, its transverse momentum, known as missing transverse momentum $\vec{p_T^{miss}}$, can be computed as the opposite of the sum of the transverse momenta of all the detected particles in the event:

$$\vec{p_T^{miss}} = -\sum_i \vec{p}_{T,i} \tag{2.13}$$

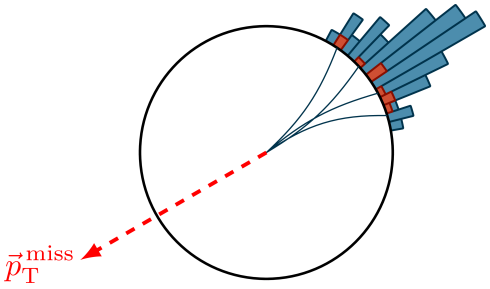


Figure 2.10 Schematic of missing transverse momentum computation.

The computed variable is shown graphically in Fig. 2.10, where the missing transverse momentum is represented as the opposite of the vectorial sum of the transverse momenta of all the detected particles in the event.
In these cases, the transverse mass is used instead of the invariant mass, which is defined as:

$$m_T = \sqrt{2p_T E_T^{miss}[1 - \cos(\Delta\phi)]} \tag{2.14}$$

where E_T^{miss} is the magnitude of the missing transverse momentum and $\Delta\phi$ is the azimuthal angle between the transverse momentum of the detected particle and the missing transverse momentum.

2.3.2 Magnetic system

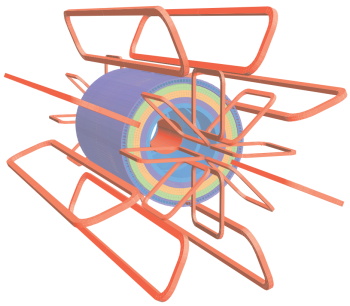


Figure 2.11 ATLAS magnetic system. Figure form Ref. [35].

Magnetic fields play a crucial role in the ALTAS detector, allowing the tracking of charged particles and the measurement of their momenta via the sagitta method [36]. The ATLAS detector relies on a hybrid superconducting magnet system composed of four sub-components: a central solenoid, a barrel toroid and two end-cap toroids [35]. This complex system was adopted to reduce particle interactions with inactive material and, more importantly, to obtain a higher momentum resolution. Cooled to 4.5 K using liquid helium, the system covers a total volume of 26 m in length and 21 m in outer diameter. A schematic of the ATLAS magnetic system is shown in Fig. 2.11.
The central solenoid is positioned inside the calorimetry system directly around the Inner Detector cavity, where it generates a highly uniform 2 T axial magnetic field

parallel to the beam pipe deflecting charged particles in the transverse plane. It measures 5.3 m in length, 19 cm in thickness and an inner diameter of 2.44 m. To minimize dead material in front of the Electromagnetic Calorimeter, the solenoid shares a single cryostat vessel and supportive cylinder with the barrel ECal.
Providing the deflection field for the outer Muon Spectrometer, the barrel toroid is composed of eight large rectangular coils mounted radially with eight-fold azimuthal symmetry around the calorimeters. Extending 25.3 m in length with inner and outer diameters of 9.4 m and 20.1 m, respectively, it generates a peak magnetic field 3.9 T. Two smaller end-cap toroid structures, 5 m long with an inner and an outer diameter of 1.65 m and 10.7 m respectively, are inserted into each end of the barrel to extend coverage into forward regions. They generate a peak field 4.1 T. Both the barrel and end-cap toroids bend charged particles in the longitudinal plane.

2.3.3 Particle tracking

As mentioned in the previous section, magnets are fundamental to measure the momentum of charged particles. In fact, when a charged particle enters a magnetic field, it is deflected by the Lorentz force, leading to a curved trajectory, whose radius of curvature is proportional to the momentum of the particle. By measuring the curvature of the track, the momentum can be inferred.
The method used to measure the particle momentum is the sagitta method [36], where the sagitta s is defined as the distance between the midpoint of the chord connecting two points on the track and the track itself. The sagitta is related to the radius of curvature R, the length of the chord L and the angle θ under the arc by the following equation:

$$s = R\left[1 - cos\left(\frac{\theta}{2}\right)\right] \simeq \frac{L^2}{8R} \tag{2.15}$$

where the assumption of small angles was used in the final step.
Making use of the relation 2.2 between the radius of curvature R, the momentum p and the magnetic field B, the sagitta can be expressed as:

$$p = \frac{0.3BL^2}{8s} \tag{2.16}$$

As direct consequence of this latter equation, the reltive uncertainty on the momentum measurement is equal to the relative uncertainty on the sagitta measurement, $\frac{\Delta p}{p} = \frac{\Delta s}{s}$, which, using σ_x position uncertainty and N number of measured points, leads to:

$$\frac{\sigma_p}{p} = \frac{\sigma_x[m]}{0.3BL^2[Tm^2]}\sqrt{\frac{720}{N+5}}p[GeV] \tag{2.17}$$

From Eq. 2.17 it is evident that the momentum resolution can improve with a stronger magnetic field, a longer track length and a higher number of measured points but it worsens with higher momentum particles since $\frac{\sigma_p}{p} \propto p$.

2.3.4 The Inner Detector

The Inner Detector (ID) [37] is the closest system to the interaction point. Its function is to reconstruct the trajectories of charged particles, measure their transverse momentum and identify both primary and secondary interaction vertices of decays. Covering a pseudo-rapidity region of $|\eta| \leq 2.5$, 7.02 m long and 2.30 in diameter, it is inserted in the central solenoid (Sec. 2.3.2), providing the tracking sub-systems an external magnetic field of 2 T parallel to the beam axis. In order to balance both optimal resolution, material budget and radiation exposure, the ID utilizes three concentric silicon-based tracking technologies organized radially outward from the beam pipe: the Pixel Detector, the Semiconductor Tracker (SCT) and the Transition Radiation Tracker (TRT). In Fig. 2.12 a schematic of the ID is shown.

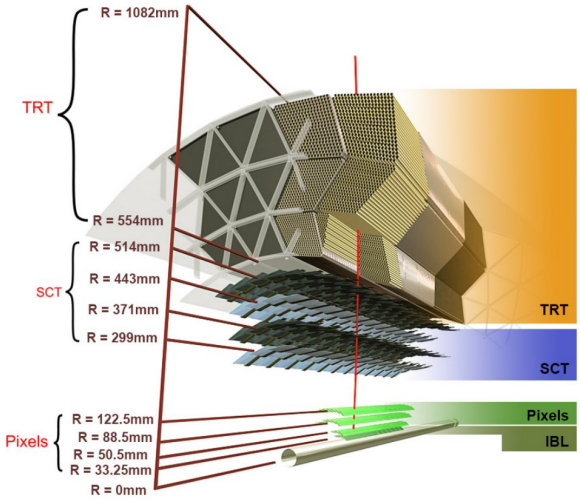


Figure 2.12 3D representation of the ATLAS inner detector. Figure form Ref. [38].

Pixel Detector

The closest system to the beam pipe is the Pixel Detector. It is composed of three pixel layers placed at distance of 50.5 mm, 88.5 mm and 122.5 mm from the IP respectively in the barrel. In the end-caps, there are installed three disks on each side located between $z = \pm 495$ mm and ± 650 mm. The pixel system has the finest granurality, delivering a spatial resolution of 10 μm in the (r, ϕ) plane and 115 μm along the longitudinal z-axis.
During LS1, an additional innermost layer was added at $r = 33.25$ mm. It is called the Insertable B-Layer (IBL) and it was added to improve the tracking performance, providing coverage up to $|\eta| = 2.58$ and to increase the radiation resistence of the ID.

SemiConductor Tracker

Outside the Pixel Detector, the SemiConductor Tracker (SCT) is positioned. It is composed of four barrel layers at distances between 299 mm and 514 mm and 18 overall end-cap disks. SCT delivers a spatial resolution of 16 μm in the (r, ϕ) plane

and 580 μm along the longitudinal z-axis.

Transition Radiation Tracker

The outermost region of the ID is composed of the Transition Radiation Tracker (TRT) situated between 554 mm and 1082 mm from the IP. Unlike the inner silicon-based detectors, the TRT consists of approximately 300,000 polyimide drift tubes, 4 mm each in diameter and equipped with a central gold-plated tungsten wire. The tubes are filled with a gas mixture based on Xenon-CO_2-O_2 (70%-27%-3%). TRT provides a single-hit spatial resolution of 130 μm in the barrel region with $|\eta| < 0.7$. In the end-caps, with $0.7 \leq |\eta| \leq 2.0$ the structure of the tubes is the one of disk, as for the other systems. The main feature of TRT is to provide continuous tracking by recording an average of 36 hits per charged track. Additionally, it is able to distinguish electrons from other objects by detecting the transition radiation photons which are emitted when charged particles cross the active materials inside the tubes.

2.3.5 Calorimeters

Calorimeters are used in ATLAS to measure the energy of particles by means of a destructive measurement on the particles. For this reason, calorimeters in ATLAS are placed outside the Inner Detector, allowing a complete reconstruction of the particle trajectory before it is absorbed [39].
Covering a region up to $|\eta| < 4.9$, the calorimetry system features sampling technology with dense absorbers alternated with active media to completely contain electromagnetic and hadronic showers. The idea behind this technique is to have a low probability of signals in the muon spectrometer from a particle different from a muon, which is the only particle that should cross the calorimeter without being absorbed (apart from not interacting particle such as neutrinos).
Liquid argon (LAr) serves as the active material, while being kept at a temperature of 88.5 K by three cryostats. When a charged particle crosses LAr, light is produced, which is collected by wavelength-shifting fibers and routed to photo-multipliers (PMTs) for the conversion to signal.
The calorimeter complex in ATLAS is divided into two components: the Electromagnetic Calorimeter (ECal) and the Hadronic Calorimeter (HCal). A schematic of the complex can be seen in Fig. 2.13.

Electromagnetic Calorimeter

The LAr ECal is composed of lead (Pb) absorbers (passive material) separated by LAr gaps (active material) with an accordion-shaped geometry. This particular layout is chosen to obtain high sampling rate while keeping a fast signal extraction time, ensuring uniform cover in ϕ. As previous systems, also ECal has a symmetric structure, with three main components: the barrel, with coverage $|\eta| \leq 1.475$ and the two end-caps in the range $1.35 \leq |\eta| \leq 3.2$. The region $1.37 \leq |\eta| \leq 1.52$ is known as *crack-region*, given the large amount of inactive material present in this area, which leads to a degraded energy resolution.

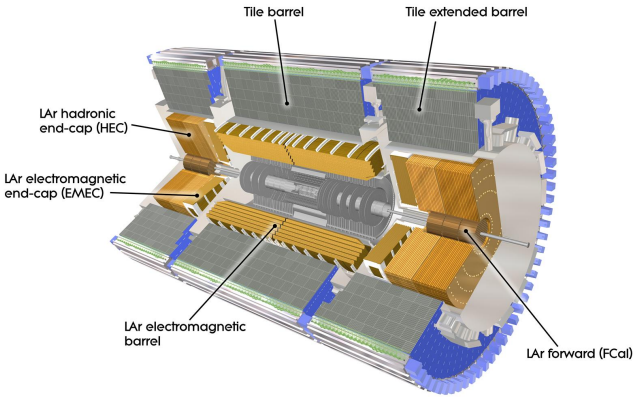


Figure 2.13 Schematic of the ATLAS calorimeter system. Figure form Ref. [35].

Based on the η region, the thickness of the ECal varies between 22 to 33 X_0, which is sufficent to contain the full electromagnetic showers.
The ECal is structured in three layers in the radial direction, each with a different granularity in η and ϕ. The first layer has a fine granularity in η, enabling the discrimination between showers coming from electrons, photons or neutral pions. The second layer collects the majority of the electromagnetic shower energy, with fine granulality in both coordinates. The third and final layer measures high-energy shower tails.
The ECal energy resolution has been measured during test-beams and, as a function of the beam energy, it is computed as:

$$\frac{\sigma_E}{E} = \frac{10.1\%}{\sqrt{E[\text{GeV}]}} \oplus 0.17\% \tag{2.18}$$

where the first term is the stochastic term, dominant at low energies, summed in quadrature with the term accounting for the non-uniform response of the detector.
A schematic representation of the ATLAS ECal is shown in Fig. 2.14.

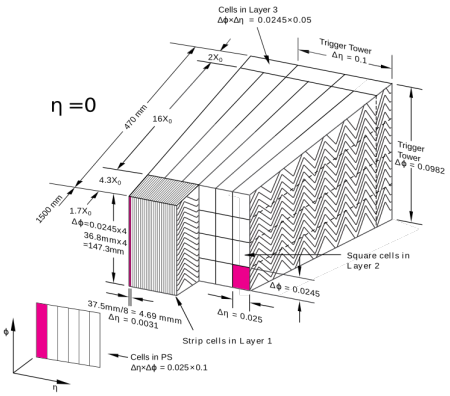


Figure 2.14 Schematic view of the accordion shape of the ATLAS ECal at $\eta = 0$. Figure form Ref. [40].

Hadronic Calorimeter

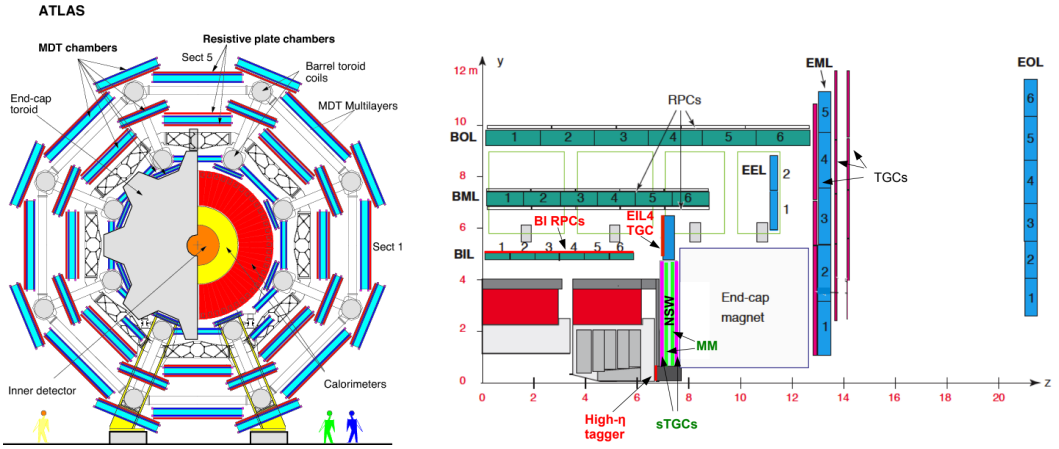
Hadronic energy measurements are handlead by the HCal, which is composed of three sub-sections: the Tile calorimeter, the Hadronic End-Cap Calorimeter (HEC) and the Forward Calorimeter (FCal). The Tile calorimeter is a sampling calorimeter with steel as absorber and scintillating tiles as active material, covering the central region of the detector with $|\eta| \leq 1.7$. With an inner diameter of 4.46 m and a outer one of 8.50 m, it covers 7.4 λ (the interaction length) at $\eta = 0$.
The HEC is based on two wheels per end-cap, with copper as absorber material covering up to 5.7 λ. With an outer diameter of 4.06 m, it covers the intermediate region of the detector with $1.5 \leq |\eta| \leq 3.2$.
In conclusion, the FCal covers the region $3.1 \leq |\eta| \leq 4.9$ and it is the closest component of the calorimeter system to the beam pipe. It is composed of three modules (FCal1, FCal2, FCal3), with copper as absorber for the FCal1, reaching 27.6 X_0, and tungsten for the other two, which allows to reach 90 X_0. The FCal is designed to resist to high radiation doses, being exposed to a high flux of particles, since a big fraction of collisions is associated with low momentum transfer.
The overall hadronic calorimeter resolution can be parametrized as:

$$\frac{\sigma_E}{E} = \frac{52\%}{\sqrt{E[\text{GeV}]}} \oplus \frac{1.6 \text{ GeV}}{E[\text{GeV}]} \oplus 3.0\% \tag{2.19}$$

where the first term, as for the ECal, is the stochastic term; the second one is associated to the noise from the read-out electronics and the last one is a constant term keeping tracks of unknown effects.

2.3.6 Muon spectrometer

The Muon Spectrometer (MS) forms the outermost layer of the ATLAS detector, with a total length of 44 m and an outer diameter of 25 m. Making use of the toroid magnets (Sec. 2.3.2) and the sagitta method (Sec. 2.3.3), it tracks particles with $p_T > 3$ GeV that are not absorbed by the different calorimeters in the region $|\eta| \leq 2.7$, which are mainly muons. Targeting a resolution of 10% for 1 TeV muon tracks, MS is required to measure sagittas down to approximately 500 μm, which is quite a challenge for the detector. As for the other ATLAS sub-detectors, the MS is organized in a barrel and in an end-cap region. The barrel covers the central region $|\eta| < 1.05$ and it consists of three concentric cylindrical shells placed at radii of 5 m, 7.5 m, and 10 m from the beam line. The two end-cap regions cover the forward zones, $1.05 < |\eta| < 2.7$ for tracking and $1.05 < |\eta| < 2.7$ for Level 1 trigger. The end-cap detectors are divided in Small Wheel, located in front of the end-cap toroids, and Big Wheel, behind them. They are mounted on large perpendicular wheels centered at longitudinal positions $|z| = 7.4$ m, 10.8 m and 21.5 m. A schematic view of the MS is shown in Fig. 2.15.
The tracking and trigger instrumentation follow a 16-fold azimuthal (ϕ) layout, matching the geometry of the toroidal magnet coils. Sub-detector stations are alternated large sectors, positioned between adjacent coils, and small sectors, covering regions directly aligned with the magnet structures. Precision tracking and



(a) (b)

Figure 2.15 Fig. 2.15a represents the transverse view of the ATLAS Muon Spectrometer. Fig. 2.15b shows the longitudinal (z, r) view of the MS, plane orthogonal to the magnetic field from the barrel coils and therefore banding plane for muons. Figure from [41].

momentum measurement rely primarily on Monitored Drift Tube (MDT) chambers. Fast triggering is taken care by Resistive Plate Chambers (RPCs).

Tracking chambers

Each MDT chamber contains aluminum drift tubes of 30 mm diameter and 400 μm thick wall, filled with an Ar:CO_2 gas mixture (93%:7%) at a pressure of 3 bar. A central tungsten-rhenium anode wire of 50 μm diameter is insterted in the tubes. It operates at 3080 V, providing a gas gain of order 10^4 and yielding a muon sagitta resolution of $\sigma_S = 45$ μm. The drift chambers are composed of multi-layers of MDTs, each one containing three or four tube layers separated by aluminum spacers. Optical alignment systems continuously monitor structural deformations to preserve the tracking resolution. Alignment represents the dominant source of uncertainty in high p_T muon measurements; with low momentum particles, the uncertainty is dominated by fluctuations in the energy deposit.
To sustain expected HL-LHC background conditions, the original Small Wheels were completely replaced for Run 3 by the New Small Wheel (NSW) system. It introduces two complementary thin-gap technologies across 16 sectors per wheel: resistive-strip MicroMegas (MMs) for precision tracking and Thin Gap Chambers (TGCs) for fast triggering. MMs modules employ a $\sim$ 130 μm amplification gap defined by a metallic micro-mesh, achieving a 100 μm spatial resolution.

Trigger chambers

In the barrel region ($|\eta| < 1.05$), doublets and triplets of Resistive Plate Chambers (RPCs) are mounted. They function as continuous gaseous detectors operating without wires, built from parallel bakelite plates separated by a 2 mm gas gap operating under an electric field of 4.9 kV/mm. RPC readout strips achieve an exceptional timing resolution of 1.5 ns and spatial resolution of 10 mm both in z

and ϕ direction.
In the forward end-caps ($1.05 < |\eta| < 2.4$), fast triggering is handled by TGCs, which are designed to work under high flux of particles. They utilize a 3.2 mm strip pitch to provide a 4 ns time resolution and a spatial resolution of 2-6 mm in the bending plane and 3-7 mm in the non-bending plane.

2.3.7 Trigger system

At the LHC, proton bunches collide every 25 ns, resulting in a collision rate of 40 MHz and a projected data occupancy of $60{\times}10^6$ MBs^{-1}. Clearly, the ATLAS detector cannot record all of these events. The ATLAS Trigger and Data Acquisition (TDAQ) system [41, 42] is designed to select interesting events for further analysis while discarding uninteresting ones, reducing the data rate by five orders of magnitude to a manageable level. This reduction relies on a two level scheme: Level-1 (L1) and High-Level Trigger (HLT). An overview of the ATLAS TDAQ system is shown in Fig. 2.16.

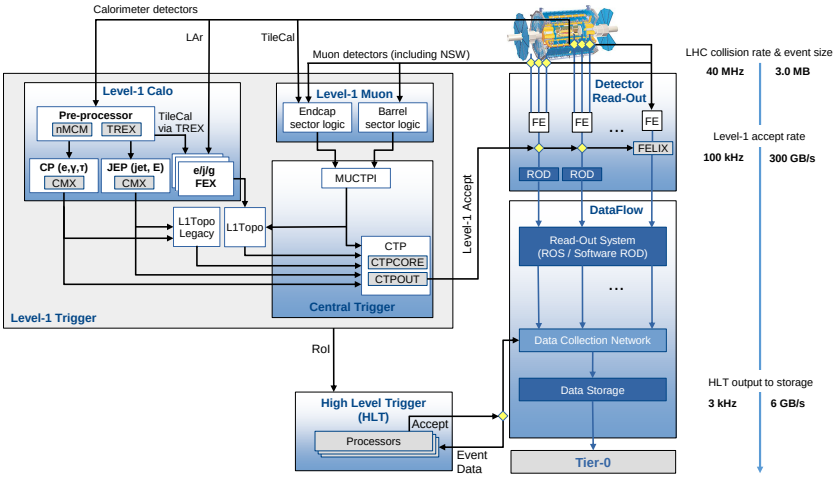


Figure 2.16 Schematic view of the ATLAS TDAQ system in Run 3. Figure form Ref. [41].

The L1 trigger, implemented using custom electronics, reduces the event rate to 100 kHz, which is the maximum rate that the ATLAS readout system can handle. It operates with a latency (time needed to take the decision) below 2.5 μs. The L1 Trigger searches for objects like electrons, photons, jets, τ-leptons, large E_T and large E_T^{miss} using reduced-granularity information from the calorimeters (L1Calo). Events containing high-p_T muons, instead, are hunted by muon system (L1Muon), composed of the RPCs and TGCs. If an event passes the L1 selection, it is sent to the HLT, together with some Regions of Interest (RoIs) identified by the L1 trigger. The HLT then processes the buffered events, operating at higher levels of detail than

the previous stage. It operates using full-granularity information from the ATLAS sub-detectors. To minimize network bandwidth, the algorithms initially focus on RoIs, defined in η and ϕ coordinates by the L1 stage. With a latency of 550 ms, the HLT reduces the event rate to approximately 3 kHz before storage.
Dedicated streams organize the stored events. While the primary physics stream retains full event information, the Trigger-Level Analysis (TLA) stream saves only partial reconstructed objects, bypassing bandwidth bottlenecks for high-rate processes. Accepted data is finally transferred to Tier-0 for offline processing.

2.3.8 The ATLAS Simulation Infrastructure

In order to interpret the data collected by the ATLAS detector and compare it with theoretical predictions, a detailed simulation of the detector response is required. The ATLAS simulation infrastructure is integrated within the Athena software framework [43], implementing the GEANT4 simulation toolkit to simulate the interaction of particles with matter [44].
This process is composed of two main steps: the *simulation* and the *digitization*. In the first one, Monte Carlo generators calculate the short-distance hard scattering, followed by parton showering, ignoring the detector geometry. After this, the generated particles with a proper lifetime $c\tau \geq 10$ mm are propagated through the detector geometry, using the GEANT4 toolkit. It simulates particles' interactions with the detector material and it records the energy deposits, together with the position and time of particles, referred as *hits*. Truth-level information is preserved throughout the process to track the exact parent-daughter decay trees.
The digitization step converts the previously cited hits into simulated electronic signals, emulating the readout electronics. To replicate realistic run conditions, the hits are combined with events due to cavern backgrounds and electronic noise. Soft QCD events are overlaid too, in order to model pile-up effects. The output of this stage are into Raw Data Objects, delivering a data format identical to the real detector readout. Consequently, identical reconstruction software used for real data are applied to simulated ones.
In conclusion, simulated events are re-weighted to match the average pile-up distribution observed in data.

Chapter 3

Hadron reconstruction in ATLAS

Before running the analysis, the signals recorded by the different ATLAS sub-detectors must be transformed into physical objects to deal with. In this chapter, the reconstruction of jets is presented, since they are the main objects of interest for the analysis conducted in this thesis.

3.1 Tracks reconstruction in ID

The ATLAS detector relies on a complex magnetic system, as described in Sec. 2.3.2, to bend the trajectories of charged particles, allowing for the measurement of their momentum. The starting point of these measurements is the reconstruction of charged particle trajectories, or *tracks*, in the Inner Detector (ID). They can be identified by groups of three-dimensional spatial points known as *hits*. Given the magnetic field configuration, tracks in the ID follow helicoidal trajectories, which can be described by the parameters: the transverse momentum p_T, the azimuthal angle ϕ, the polar angle θ, the impact parameter d_0, the longitudinal impact parameter z_0 and q/p, the momentum dividing the charge. Those parameters are graphically represented in Fig. 3.1.

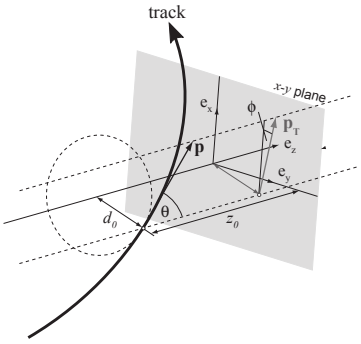


Figure 3.1 Parameters describing a track in the ATLAS detector. Figure form Ref. [45].

The tracks are reconstructed using an *inside-out tracking* strategy, starting from the innermost layers of the ID and moving outwards, with the requirement of at least 8 hits and $p_{\mathrm{T}} > 500$ MeV . The reconstruction algorithm begins by identifying clusters of hits in both the Pixel and SCT sub-detectors. Those clusters then form *track seeds*, which are composed by triplets of them. Using a combinatorial Kalman filter, the track seeds are extended into full track candidates, on which a global χ^2 minimization fit is performed to determine the initial trajectory parameters. After adding compatible hits from the TRT sub-detector, if the requirements are satisfied, another fit with higher precision is performed to refine the track parameters.
The tracks which have been reconstructed as described above, are then stored and used to reconstruct the primary interaction vertex (PV). Among all vertices, the one with the largest scalar sum of squared track transverse momenta, $\sum p_{\mathrm{T}}^2$, is identified as the vertex of the hard scattering (HS) and its position is taken as reference point to reconstruct all the other objects.

3.2 Jets reconstruction

As described in Sec. 2.2, the scattering of two partons in a *pp* collision produces high-energy quarks and gluons, which undergo a process of parton showering and hadronization, producing collimated sprays of particles, which are known as *jets*. A schematic of the jets nature is presented in Fig. 3.2.

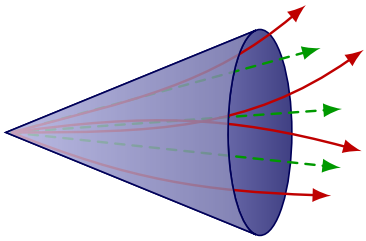


Figure 3.2 Drawing of a jet, showing the cone of particles produced by the hadronization of a parton.

Jets costituents then form hadrons due to the QCD colour confinement, which are detected by the ATLAS sub-detectors.
The reconstruction of jets is a complex process, which requires the combination of information from different sub-detectors. The diffent components of the spray are detected and reconstructed as different objects, which are then combined to form the final jet object. The combination is the not trivial part and dedicated algorithms have been developed to perform it.

3.2.1 Jet constituents reconstruction

The reconstruction of jets starts with the reconstruction of their constituents, which are the physical objects detected by the ATLAS sub-detectors. This process uses different algorithms depending on the target physics regime and kinematic regime,

combining the measurements of the ID with those from calorimeters. The common starting point is the clustering of calorimeter cells, which are grouped into *topological clusters*.

Topoclusters

Topoclusters, or jet topological clusters, are three-dimensional clusters of energy deposits in calorimeter cells. They dynamically adapt their spatial volume based on the topology of the energy shower across both the LAr and Tile calorimeter layers. The formation of topoclusters is controlled by the parameter ζ_{cell}, the cell signal significance[46], computed as the ratio between the signal E_{cell} and the average noise $\sigma_{noise,cell}$ in the cell:

$$\zeta_{cell} = \frac{E_{cell}}{\sigma_{noise,cell}} \tag{3.1}$$

All the cells with $\zeta_{cell} \geq 4$ are considered as seeds for the topocluster formation. Starting from these seed cells, the cluster is grown by adding adjacent cells with lower significance, $\zeta_{cell} \geq 2$. Finally, all the cells with $\zeta_{cell} \geq 0$ are added to the cluster if they are adjacent to a cell already added to the cluster.
In the case of overlapping particle showers, the topocluster can contain multiple local energy maxima, which are then split into two separate clusters, with the requirement $E_{\text{cell}} > 500$ MeV.
To each topocluster is assigned a four-momentum P_{clus} and it is treated as a pseudo-particle. In fact the mass is set to zero, since there is no information about the mass of the particles producing the energy deposits, and the energy is computed as the sum of the energies of the cells in the cluster:

$$E_{\text{clus}} = \sum_i E_{\text{cell},i} \tag{3.2}$$

The direction parameters of the cluster, η_{clus}, ϕ_{clus}, are determined from the energy-weighted barycenter of the cluster relative to the nominal origin of the ATLAS coordinate system:

$$\eta_{\text{clus}} = \frac{\sum_i |E_{\text{cell},i}|\eta_{\text{cell},i}}{\sum_i |E_{\text{cell},i}|}, \qquad \phi_{\text{clus}} = \frac{\sum_i |E_{\text{cell},i}|\phi_{\text{cell},i}}{\sum_i |E_{\text{cell},i}|} \tag{3.3}$$

Absolute cell energies are utilized in the denominator weighting to prevent directional bias from negative cell energy fluctuations induced by noise or pile-up subtraction. Those kinematic variables are used to distinguish between electromagnetic and hadronic showers. In fact, as described in Sec. 2.2.3, hadronic showers contain also EM components, resulting in the fact that the reconstructed clusters are only partially hadronic. For this reason, after Local Cell Weighting (LCW) calibrations of the topoclusters, a probability $P_{\text{EM}}^{\text{clus}}$ is assigned. It tells us the probabilty that the deposit is electromagnetic versus hadronic. Topoclusters with $P_{\text{EM}}^{\text{clus}} < 0.5$ are identified as hadronic, otherwise they are identified as electromagnetic.

Particle Flow (PFlow)

The Particle Flow (PFlow) algorithm [47] is built to optimize the reconstruction of jets with lower transverse momentum, 0.5 GeV $\leq p_T \leq$ 40 GeV. This algorithm relies on the fact that at lower energies, the momentum resolution is higher than the energy one. This feature is evident in the formulas associated with the two resolutions, Eq. 2.17 and Eq. 2.19, which show the different scalings with respect to the momentum ($\sigma_{p_T}/p_T \propto p_T$) and energy ($\sigma_E/E \propto 1/\sqrt{E}$) respectively.
The algorithm begins by the selection of the tracks to target: 0.5 GeV $\leq p_T \leq$ 40 GeV, $|\eta| < 2.5$, at least 9 silicon hits, zero pixel holes and $E_{clus}/p_{trk} > 0.1$ (E_{clus} is the energy of the matched topocluster and p_{trk} is the momentum of the track). After this, PFlow subtracts from the matched topoclusters the expected energy deposit from charged hadrons of the selected tracks, replacing it with the expected energy deposit estimated from the track momentum p_{trk}, ensuring higher resolution. If the residual energy of a topocluster after subtraction is consistent with fluctuations expected from a single-particle signal, the remnant cluster is completely removed. If significant energy remains, it is treated as a neutral energy remnant.
The finalized PFlow constituents consist of selected charged tracks for charged hadrons and un-subtracted or remnant topoclusters for neutral hadrons and photons.

Track Calo-Clusters

In highly boosted regimes ($p_{\mathrm{T}} > 200$ GeV), the angular separation between hadronic decay products shrinks below the spatial resolution of individual calorimeter cells [**?**]. Particle Flow performance degrades due to high track density and calorimeter energy overlap [**?**]. Track-CaloClusters (TCCs) circumvent this limitation by combining the track directional precision evaluated at the perigee with topocluster energy measurements [**? ?**].

Matching and Sharing Scheme Tracks from the primary vertex are extrapolated to the calorimeter and matched to topoclusters if their relative angular separation satisfies [**?**]:

$$\Delta R < \sqrt{\sigma^2_{\mathrm{clus}} + \sigma^2_{\mathrm{trk}}} \tag{3.4}$$

where σ_{clus} is the topocluster width and σ_{trk} represents the track extrapolation uncertainty [**?**]. Unlike PFlow, TCC does not perform cell-level energy subtraction [**? ?**]. Instead, it redistributes cluster energies according to four distinct categories [**? ?**]:

- **Single Track Matched to Single Cluster (Combined TCC):** A single object is formed taking its energy E from the topocluster and its direction (η, ϕ) from the track [**? ?**].
- **Unmatched Topocluster (Neutral TCC):** Originates from neutral hadrons (K^0_L, n, γ) [**? ?**]. Kinematic parameters (E, η, ϕ) are taken directly from the LCW topocluster [**? ?**].

- **Unmatched Track (Charged TCC):** Represents a charged particle that deposited negligible energy in the calorimeter [**?** **?**]. The object uses the track momentum and direction [**?** **?**].
- **Multiple Tracks Matched to One or More Clusters:** One TCC object is created per track, setting its direction to the track coordinates [**?** **?**]. The total energy of the matched topocluster group is split among the tracks proportionally to their measured track transverse momenta $p_{\mathrm{T,\,trk}}$ [**?** **?**].

TCC objects preserve fine angular substructure within large-radius ($R = 1.0$) jets, improving jet mass and substructure observable resolutions (τ_{21}, D_2) in high-p_{T} vector boson tagging [**?** **?**].

Unified Flow Objects

While PFlow excels at low p_{T} and TCC dominates at high boost, utilizing separate constituent definitions across different analysis channels introduces reconstruction inconsistencies [**?**]. Unified Flow Objects (UFO) synthesize the PFlow and TCC methodologies into a single unified input definition [**?**].

UFO Construction Workflow

1. **PFlow Baseline & CHS:** The standard PFlow algorithm is executed [**?**]. Charged PFlow objects associated with pile-up interaction vertices are removed via CHS [**?**].
2. **Neutral Subtraction:** Advanced pile-up density mitigation algorithms, such as Constituent Subtraction or SoftKiller, are applied to the remaining neutral PFlow topoclusters to remove uncorrelated soft radiation [**?**].
3. **TCC Resampling Algorithm:** The surviving neutral PFlow clusters and hard-scatter ID tracks that were not subtracted during the initial PFlow step are passed into the TCC cluster-splitting algorithm [**?**].
4. **UFO Output:** In low-p_{T} environments, UFO objects default to standard CHS PFlow performance due to complete PFlow energy subtraction [**?**]. In high-p_{T} boosted environments where PFlow leaves un-subtracted energy remnants, the TCC re-clustering engine splits neutral energy based on track directions [**?**].

By integrating PFlow subtraction at low p_{T} with TCC angular splitting at high p_{T}, UFO inputs optimize small-radius ($R = 0.4$) and large-radius ($R = 1.0$) jet performance simultaneously across the full momentum spectrum [**?**].

3.2.2 Jet algorithms

3.2.3 Detailed Jet Reconstruction and Substructure Variables

Jet Constituent Definitions and Clustering

ATLAS constructs jets from multiple classes of input objects depending on the target physics regime and kinematic environment [**?**]:

- **Particle Flow (PFlow):** Combines Inner Detector tracking with calorimeter topological clusters (topoclusters) [**?** **?**]. The PFlow algorithm evaluates the expected energy deposition of charged tracks in the calorimeter layers and subtracts it cell-by-cell [**?** **?**]. Retaining track momenta for low-p_{T} charged hadrons and neutral topoclusters yields superior p_{T} and angular resolution while suppressing pile-up contamination [**?** **?**].

- **Track-CaloClusters (TCC):** Designed primarily for high-p_{T} large-radius jet reconstruction [**?**]. TCC objects match the directional parameters of inner detector tracks to calorimeter topoclusters, leveraging tracking precision to regain angular resolution in densely collimated particle environments [**?**].

- **Variable-Radius (VR) Track Jets:** Built by clustering high-quality Inner Detector tracks ($p_{\mathrm{T}} > 500$ MeV) associated with the primary vertex [**?**]. VR jets employ a dynamic radius parameter scaled inversely with track p_{T} according to $R_{\mathrm{eff}}(p_{\mathrm{T}}) = \min(0.4, \max(0.02, \rho/p_{\mathrm{T}}))$ with $\rho = 30$ GeV, allowing efficient decay product capture across wide momentum spans [**?**].

Constituent objects are combined using sequential recombination algorithms [**?** **?**]. The distance d_{ij} between entities i and j, alongside the beam-distance d_{iB}, are defined as [**?** **?**]:

$$d_{ij} = \min\left(p_{\mathrm{T},i}^{2p}, p_{\mathrm{T},j}^{2p}\right)\frac{\Delta R_{ij}^2}{R^2}, \qquad d_{iB} = p_{\mathrm{T},i}^{2p} \tag{3.5}$$

where $\Delta R_{ij}^2 = (y_i - y_j)^2 + (\phi_i - \phi_j)^2$, R is the distance parameter, and p dictates the clustering metric [**?** **?**]. The default choice across ATLAS is the anti-k_t algorithm ($p = -1$), which generates regular, cone-like jet boundaries around hard constituent seeds while minimizing susceptibility to soft background radiation [**?** **?**]. Standard configurations adopt $R = 0.4$ for resolved small-radius jets and $R = 1.0$ for boosted large-radius jets [**?** **?**].

Jet Grooming and Calibration

Large-radius jets ($R = 1.0$) inherently collect non-negligible soft energy contributions arising from the underlying event, initial-state radiation, and in-time or out-of-time pile-up interactions [**?** **?**]. Grooming algorithms remove these uncorrelated soft components [**?** **?**].

ATLAS heavily utilizes the *trimming* algorithm [**?** **?**]. Trimming operates by reclustering the constituent elements of a large-radius jet into smaller subjets using the k_t or Cambridge/Aachen algorithm with a smaller sub-radius $R_{\mathrm{sub}} = 0.2$ [**?** **?**]. Any subjet failing a relative momentum fraction threshold f_{cut}, defined as:

$$\frac{p_{\mathrm{T},i}^{\mathrm{subjet}}}{p_{\mathrm{T}}^{\mathrm{jet}}} < f_{\mathrm{cut}} \qquad (\text{typically } f_{\mathrm{cut}} = 0.05) \tag{3.6}$$

is discarded from the final clustered object [**?** **?**].

To ensure accurate energy and mass scale measurements, a sequential calibration workflow is implemented [**?**]. Jet Energy Scale (JES) corrections account for

calorimeter non-compensation, inactive detector material, and pile-up offsets using event-by-event area-based density estimations (ρ) combined with *in-situ* balance measurements (Z/γ + jet and multijet events) [**?**]. For large-radius jets, Jet Mass Scale (JMS) calibrations restore accurate invariant mass distributions using track-assisted mass combinations and *in-situ* forward-folding fits in $t\bar{t}$ events [**?**]. Furthermore, the Jet Vertex Tagger (JVT)—a multivariate discriminant combining track-vertex association parameters—is applied to small-radius jets to reject pile-up artifacts [**?**].

Jet Substructure Observables

When massive unstable particles (e.g., W, Z, H bosons or top quarks) decay hadronically with high transverse momentum ($p_{\mathrm{T}} \gg m$), their decay products become Lorentz-boosted into a single large-radius jet envelope [**?** **?**]. Identifying such boosted topologies against single-parton QCD background jets relies on characterizing energy distributions within the jet core through substructure variables [**?** **?**].

N-Subjettiness (τ_N) N-subjettiness quantifies the degree to which a jet's constituent energy is aligned along N candidate subjet axes [**?** **?**]. It is calculated as [**?** **?**]:

$$\tau_N = \frac{1}{R_0 \sum_i p_{\mathrm{T},i}} \sum_i p_{\mathrm{T},i} \min\left\{\Delta R_{1,i}, \Delta R_{2,i}, \ldots, \Delta R_{N,i}\right\} \tag{3.7}$$

where i iterates over all jet constituents, R_0 is the main jet radius, and $\Delta R_{k,i}$ represents the spatial distance between constituent i and subjet axis k (typically derived via exclusive k_t or winner-takes-all axes) [**?** **?**]. Small values of τ_N indicate that the radiation is concentrated around N or fewer cores [**?**]. Disambiguation of N-prong hypotheses is achieved using ratio variables:

$$\tau_{N,N-1} = \frac{\tau_N}{\tau_{N-1}} \tag{3.8}$$

The ratio $\tau_{21} = \tau_2/\tau_1$ effectively isolates two-prong vector boson decays ($W/Z \to q\bar{q}$) from single-prong QCD jets, whereas $\tau_{32} = \tau_3/\tau_2$ discriminates three-prong top-quark decays ($t \to bW \to bq\bar{q}'$) [**?** **?**].

Energy Correlation Functions (ECFs) and D_2 Energy Correlation Functions measure multi-particle angular correlations without necessitating candidate subjet axis definitions [**?** **?**]. The v-point correlation functions for a jet J are expressed as [**?** **?**]:

$$\mathrm{ECF}_1 = \sum_{i \in J} p_{\mathrm{T},i} \tag{3.9}$$

$$\mathrm{ECF}_2(\beta) = \sum_{i<j \in J} p_{\mathrm{T},i} p_{\mathrm{T},j} (\Delta R_{ij})^{\beta} \tag{3.10}$$

$$\mathrm{ECF}_3(\beta) = \sum_{i<j<k \in J} p_{\mathrm{T},i} p_{\mathrm{T},j} p_{\mathrm{T},k} (\Delta R_{ij} \Delta R_{ik} \Delta R_{jk})^{\beta} \tag{3.11}$$

where β is an angular weighting exponent [? ?]. Combining these function topologies yields the dimensionless discriminant D_2, defined as [? ?]:

$$D_2(\beta) = \frac{\mathrm{ECF}_3 \cdot (\mathrm{ECF}_1)^3}{(\mathrm{ECF}_2)^3} \tag{3.12}$$

Low D_2 values strongly correlate with two-hard-prong radiation profiles, serving as a powerful handle to isolate boosted hadronically decaying resonances [? ?].

3.3 Version 2

3.3.1 Hadronic Jet Reconstruction and Calibration

Hadronic jets represent collimated sprays of particles from partonic fragmentation and hadronization [? ?]. Jets are clustered using the sequential recombination anti-k_t algorithm, choosing a radius parameter of $R = 0.4$ for narrow-radius jets or $R = 1.0$ for large-radius boosted decay products [? ?]. Input constituents are constructed using different strategies to improve angular and energy resolution. Topological clusters calibrated at the electromagnetic scale define the baseline EMTopo jets [?]. To improve low-momentum resolution, the Particle Flow algorithm matches Inner Detector tracks to calorimeter topo-clusters, subtracting the expected charged-hadron energy deposits cell-by-cell and replacing them with the superior track momentum measurements [? ?]. At high transverse momentum, Track-CaloClusters combine the angular parameters of Inner Detector tracks with the energy measurements of topo-clusters to resolve dense shower structures [? ?]. Unified Flow Objects integrate Particle Flow and Track-CaloClusters, maintaining performance across all energy regimes [?]. Large-radius jets are groomed via trimming to suppress pile-up and soft underlying event contributions by removing low-momentum subjets [?]. Calibration sequences scale the raw jet energy to the truth-particle level [? ?]. First, pile-up offset corrections subtract the average energy density on an event-by-event basis [? ?]. Absolute jet energy scale corrections are then applied using simulated templates [? ?]. Global Sequential Calibration refines the response using jet properties like track multiplicity or longitudinal shower profiles, utilizing a neural network configuration during Run 3 [?]. Finally, in-situ calibrations correct remaining data-to-simulation discrepancies by balancing jet transverse momentum against well-calibrated reference objects, such as Z bosons or prompt photons [? ?]. Pile-up jets are further rejected using the track-based Jet Vertex Tagger likelihood discriminant [? ?].

3.3.2 Missing Transverse Momentum

Conservation of momentum in the transverse plane allows the reconstruction of weakly interacting or stable invisible particles [? ?]. The missing transverse momentum vector, $\vec{p}_{\mathrm{T}}^{\,\mathrm{miss}}$, is defined as the negative vector sum of the transverse momenta of all reconstructed and calibrated hard physics objects, plus a soft term [? ?]. Hard objects include electrons, photons, muons, tau leptons, and jets [? ?]. The soft term accounts for signals not associated with any hard object [? ?]. To minimize pile-up contamination, the Track Soft Term is calculated using only

reconstructed Inner Detector tracks associated with the primary hard-scatter vertex, omitting neutral calorimeter deposits [**?** **?**]. Overlaps between reconstructed objects are resolved sequentially to prevent double-counting of detector signals [**?** **?**].

Chapter 4

Anomaly Detection

Chapter 5

ADJJ analysis

Chapter 6

Conclusions

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